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**Secure Peer-to-Peer Payment System on Android-Based Mobile  
Devices via Vibration and Acceleration**

**(Volume \_\_ of \_\_)**

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## Secure Peer-to-Peer Payment System on Android-Based Mobile Devices via Vibration and Acceleration

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### ABSTRACT

The rapid growth of smartphone penetration in the modern era has enabled users to use mobile payment services for everyday transactions. The recent COVID-19 pandemic has significantly increased mobile payment usage and encouraged new account creation and mobile transactions in various peer-to-peer (P2P) mobile payment systems, such as Octopus and mobile wallets. The P2P mobile payment system enables two parties to securely transact by interacting in close physical proximity with mobile devices. The close physical proximity interaction is enabled by the Radio-frequency (RF) technology through Near Field Communication (NFC) and Bluetooth modules embedded in mobile phones. However, RF technology is vulnerable to several cyberattacks, such as eavesdropping and Man-in-the-Middle (MITM) attacks. It highlights the security problems of RF technology and encourages research on alternative solutions for P2P communication. The project proposes a new radio-free P2P communication channel based on vibration for mobile payment systems. The built-in vibrator and accelerometer on the mobile phone are used to send and receive vibration data, ensuring payments are conducted within a physical proximity. Moreover, to fulfill the security goals of the mobile payment system, such as confidentiality, integrity, and availability (CIA) measures, it is essential to integrate cryptographic frameworks for protecting the communication channel during transactions. However, vibration communication channels are low-bandwidth, making them unsuitable for traditional cryptographic protocols. In this study, we propose a lightweight cryptography (LWC) framework that incorporates out-of-band key establishment via a passphrase and PBKDF2, and integrates AES-CTR encryption with HMAC-SHA256 in the communication channel. These LWC protocols aim to ensure the transaction is safe. Finally, functional, Performance, and Penetration testing of the proposed secure vibration communication channel were conducted, which shows that it is feasible in a secure P2P mobile payment system under certain circumstances. However, the system has demonstrated an alternative, secure, radio-free communication methodology that can potentially be used in further research on P2P communication and lightweight cryptography under a payment system.

**Keywords**—*P2P Payment System, Vibration-based Communication, Mobile System Security, Lightweight Cryptography*

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**When life gives you lemons, make lemonade. See you all on the next journey ☺!**

## **1. Introduction**

This section will begin with an introduction to the background and research gap. Moreover, it will also define the project objectives, justify the scope, and outline the deliverables for developing a secure Peer-to-Peer (P2P) payment system using a physical vibration channel. Lastly, it will provide an overview of the paper's organization.

### **1.1 Background of the Study**

The increasing use of smartphones has fundamentally transformed the financial structure of modern society. Peer-to-Peer (P2P) payment systems have appeared as a keystone of modern digital payment transactions, enabling money transfer between users via mobile devices directly without using cash [1]. The recent COVID-19 pandemic has increased the acceptance of mobile payments, and the P2P payment market proportions reached \$1,889.16 billion in 2020 and are valued at approximately \$9,097.06 billion by 2030 globally [2]. Currently, Hong Kong has the highest percentage (46.7%) of interested users who use mobile payment system services monthly in the Asia-Pacific [3]. P2P payment systems are expected to grow rapidly and play an essential role in digital financial services worldwide.

Radio-frequency (RF) technology has seen rapid advancements in wireless network communication and mobile technology. Wireless payments using technologies such as Near Field Communication (NFC) and Bluetooth have become popular and are a standard protocol for short-range wireless communication channels. NFC has been adopted in P2P payment systems like Apple Pay, PayPal and Octopus [4, 5]. However, the RF-based approach requires particular hardware and can be vulnerable to compromise, such as eavesdropping and relay attacks [6]. Therefore, RF's drawbacks have encouraged the use of different channels for P2P communication, such as the High-Frequency Acoustic Wave channel [7] and the Visible Light channel [8]. Mobile phone device vibration and accelerometer sensing have also been proposed as a communication channel [9, 10]. These approaches may enable an offline physical P2P payment system to function without radio frequency.

### **1.2 Problem Statement**

Previous research has successfully demonstrated the feasibility of using a smartphone's vibrator and accelerometer to establish data transfer through a physical channel [9, 10]. However, these studies have primarily focused on basic P2P communication for experimental purposes. Implementing the physical vibration channel, particularly in financial applications such as a P2P

payment system, remains largely unexplored. Few studies have examined the feasibility of the offline physical vibration-based P2P payment system. Proposing an innovative method for future research is necessary.

Moreover, deploying a security solution like an encryption scheme is crucial for protecting the P2P payment system against threats. However, the use of high-overhead traditional cryptographic protocols is not suitable in the case of vibration-based communication with severe constraints. Scholars have found that asymmetric encryption and public-key infrastructure (PKI) are particularly unsuitable. Their computational complexity drains resources, and a significant communication overhead is difficult for a low-bit-rate channel [11]. This creates a critical research gap in designing a lightweight cryptographic mechanism for high-latency, low-throughput vibration-based communication inside the P2P payment system.

### **1.3 Aim and Objectives**

This project aims to tackle the insufficiency of using the vibration channel in the offline P2P payment system and design a lightweight encryption and key exchange mechanism for this communication channel. The objectives are as follows:

- To design a Vibration-Based Communication Protocol for the system: Develop an application-based algorithm for reliably encoding, transmitting, and decoding the data between two Android smartphones using the devices' vibrator and accelerometer.
- To develop a Lightweight Cryptographic Framework: Implement a simple security mechanism for the vibration channel. These include a secure custom method for secret key exchange and a simple encryption scheme for the transaction data.
- To evaluate the system performance and security: Test the final prototype of the application, and assess its performance, such as accuracy; Analyze its security against a described threat model and identify possible vulnerabilities and limitations.

### **1.4 Project Scope and Deliverables**

This project will develop an Android application that uses only the built-in vibrator and accelerometer for communication, excluding other technologies such as Wi-Fi or Bluetooth. The project outcome will be a functional P2P payment System with the proposed communication



protocol and cryptographic framework. The scope does not include development for other operating systems (OS) and defense against advanced physical attacks.

### **1.5 Overview of the Final Report**

The rest of the paper is organized as follows. Section 2 will conduct a literature review to indicate the necessity of the work. Section 3 presents the overall design of the proposed vibration-based P2P payment system. Section 4 illustrates the detailed implementation within the system. Section 5 presents the final results and the system's evaluation. Finally, the paper concludes with a summary of achievements, a critical review, and a discussion of the system's future work in section 6.

## **2. Literature Review**

This section begins with an overview of existing P2P mobile payment system solutions, including NFC and Bluetooth, and a review and comparison of alternative solutions to address their weaknesses. Next, a review of current vibration-based communication channels will be conducted to assess their applicability and limitations for this project. Additionally, it will provide an evaluation of a cryptographic framework for vibration channels, followed by a critical examination of its limitations in the system.

### **2.1 Existing P2P Payment Technologies and Security Challenges**

#### **2.1.1 Near Field Communication (NFC) & Security Vulnerabilities**

Nowadays, most P2P payment systems adopt the use of Radio-Frequency (RF)-based technology like NFC in personal electronic wallets (e-wallets) for contactless payment [13, 14]. NFC is a short-range, P2P communication technology that combines the benefits of Radio Frequency Identification (RFID) with wireless communication technologies like Wi-Fi. NFC is being introduced as a simple operation technology, as users only need to wave the mobile device close to another device to receive NFC signals, and the transaction can be completed immediately without other actions required [15]. This has fostered convenience and flexibility by tapping mobile devices for P2P payment without typing transaction details [16]. However, the protocol introduced a susceptibility to specific vulnerabilities, such as eavesdropping and Man-In-The-Middle Attack (MITM Attack), due to dependency on continuous internet connectivity [6, 17].

Regarding eavesdropping, NFC faces significant security challenges, creating opportunities for alternative approaches. Previous research has shown that data on an NFC-enabled device can be intercepted at a distance by an attacker who listens to RF signals carrying personal information or payment details and extracts the data from received signals during a P2P payment transaction. [18, 19]. This issue highlights security and privacy risks. Data branching can lead to unauthorized access, allowing attackers to gain unpermitted access to users' financial data throughout the NFC process. This violates the security principle of availability of "unauthorized denial of use" as the attackers can use and access the information without permission [20, 21]. Moreover, the data exposure risk has enabled attackers to collect and track users' information and potentially misuse it to commit crimes or create duplicate cards containing credit card data for card skimming [22].

Concerning the MITM attack, it could occur during NFC payments. Some researchers found that some NFC devices are connected to the cloud (Internet) channel rather than the traditional NFC

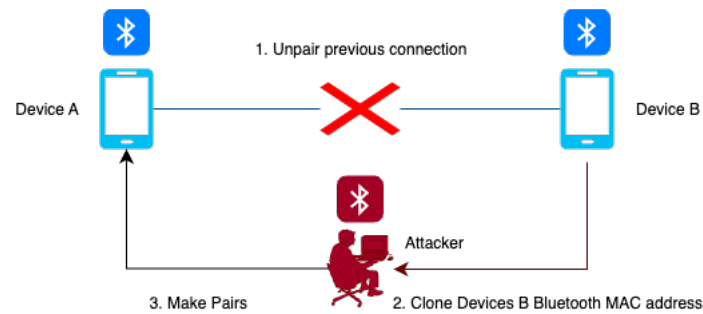
channel, where cryptographic operations occur within tamper-resistant hardware (Controller) to prevent cyber attacks [23]. They discover that if the communication channel between the mobile device and the cloud is not adequately secured, for example, with weak HTTPS protocols or unverified certificates, attackers may intercept traffic by creating untrusted CA certificates that fool users into signing them. and keep the MITM proxy for accessing the information during the transaction [23, 24].

The security loopholes of NFC highlight the need for an offline-capable payment solution that can operate without continuous internet dependency and radio-free communication protocols to avoid possible attacks.

### **2.1.2 Bluetooth & Security Vulnerabilities**

Apart from NFC, Bluetooth is also a radio-frequency (RF) technology that connects mobile devices via P2P communication. Currently, the latest standard available is versions 5 and 6 with Bluetooth Low Energy (BLE), which help reduce power consumption while maintaining short-distance communication [25]. Bluetooth operates on radio frequencies in the 2.45 GHz band, with a transfer rate from 1Mbps to 2 Mbps, and a transmission range from 10m to around 240m in omnidirectional format [25, 26]. Some researchers have implemented a “P2P-Paid system” that successfully conducts P2P mobile payments on mobile phones via a nearby Bluetooth wireless network, demonstrating the feasibility of providing a functional P2P payment system using Bluetooth [4]. However, like NFC, Bluetooth relies on an RF signal, which raises security concerns about Confidentiality, Integrity, and Availability (CIA) [27].

Regarding the Bluetooth security issues, previous research found that Bluetooth-enabled attacks are prone to security problems. For example, a Bluetooth sniffer can intercept the raw data exchanged between two mobile devices [28], which could compromise the confidentiality of transaction information, as third parties may collect sensitive data without notice. Furthermore, *Figure 2.1* shows that a MAC Spoofing attack can be made before the Bluetooth pairing between mobile devices. The attackers can clone a MAC address of the original device and impersonate it to manipulate data while in transition, showing that the information is being altered to confuse the receiver, violating the integrity [27, 29].



*Figure 2.1: A MAC spoofing attack scenario using Bluetooth*

Additionally, Bluetooth transmission must occur as RF waves. An attacker introduces sufficient RF noise into the Bluetooth network. Unwilling echoes can overwhelm the connection and prevent the establishment of communication [28]. This represents a Denial-of-Service (DoS) attack that blocks users from accessing a service by making it unavailable in a timely manner, violating the availability principle in security [29].

The security inadequacies of Bluetooth emphasize the ongoing challenges in wireless P2P payment technology. These limitations create opportunities for a more secure and offline option, a P2P communication protocol in a payment system, like vibration-based solutions.

### **2.1.3 Secure Offline Covert P2P Communication technologies**

To overcome the limitations of NFC and Bluetooth, research on alternative solutions such as physical covert P2P communication channels has demonstrated that they offer better physical security. The radio-free method needs close physical contact between mobile devices making it resistant to wireless attacks, such as eavesdropping [10]. Additionally, the lack of address discovery via the network or devices, as well as the short communication range, further safeguards data transfer [30]. Currently, there are some covert channels, for example Visual Light Communication (VLC) with Light Emitting Diodes (LED) approach proposed by Schmid et al. [8], where the approach sends information by a standard LED driven by a designed modulator that changes light intensity at a data rate to send out messages, and the receiver operates the LED in a photodetection mode that converts detected light into a photocurrent and decode it to a message. Additionally, Li et al. have proposed a communication channel based on high-frequency acoustic waves [7]. In which the sender transmits high-frequency sound wave data through the air, and the receiver captures the audio and decodes the data. Moreover, many scholars have proposed the vibration communication channels [9, 10, 31, 33, 34, 35]. The data transmission sends a vibration pattern that encodes data using the smartphone's built-in vibrator, which the receiver decodes

using the smartphone's accelerometer. In the context of the P2P mobile payment system, vibration is a more suitable covert channel than others. Compared to the LED approaches, the LED requires additional hardware modules such as an Arduino board and LEDs embedded in the mobile device, which can be a burden for carrying and not user-friendly for mobile users, and may also highly interfere with environmental factors (Such as sunlight or indoor lighting) during transactions [8]. Compared with the acoustic approach, although acoustic signals can be implemented through built-in smartphones' microphones and speakers, from an interference perspective, vibrations are less vulnerable to ambient noise, as busy or even quiet acoustic environments may degrade the sound channel [7]. The vibration on a table is relatively isolated from noise and light, showing that the channel is separated from the background environment, and it is a user-observable channel with strong physical proximity binding that helps reduce the risk of conversion attacks [9, 10].

Therefore, to achieve reliability and secure payments without connectivity, this project proposed a sustainable vibration-based P2P transaction solution that ensures financial inclusion for those concerned about payment privacy and security.

## **2.2 Vibration-Based P2P Communication Technology**

### **2.2.1 Vibration and Acceleration Data Transfer in Mobile Devices**

Vibration-based P2P Communication is a non-radio-based, low-rate, short-range communication that enables users to transfer data between smartphones by simply placing one smartphone on another, similar to NFC communication [10, 31]. The Communication requires the raw sensor data of the accelerometers and vibrators for the information sender and receiver, and this is feasible since this type of motion sensor is provided by the Android Hardware [32]. Data transmission uses a built-in vibrator in the smartphone to send a vibration pattern that the smartphone's accelerometer decodes as a receiver. Current Vibration Communication methodologies adopt a similar decode mechanism using simple binary and different adaptive thresholding; however, they differ in the encoding phase [33]. There are three primary encoding methodologies: An On-Off Keying (OOK)-based protocol [10], an Amplitude modulation (AM)-based protocol [34], and a Frequency Modulation (FM)-based protocol [35]. Previous research shows that OOK is considered more practical when applied to mobile devices, as it is simple to implement in all mobile phones by just switching the vibrator on/off to represent 1s and 0s using basic motors and API without considering the vibrator strength or frequency that may require additional hardware or OS support [33]. Therefore, OOK is the most suitable method for mobile devices, given its security and development convenience.

## **2.2.2 Applications and Limitations in Previous Approaches**

In the context of P2P communication, several research projects use the mobile phone vibrator and accelerometer for data transmission. Hwang et al. proposed VibComm, an offline Android application that uses vibration-based communication to exchange data between multiple users [9]. Yonezawa et al. have also developed Vinteraction, an information-transmitting application that leverages a combination of an accelerometer and vibrator using smartphones [31]. Both previous studies used vibration for P2P communication with OOK encoding and a similar adaptive thresholding decoding mechanism, and both were offline and radio-free. Results found that 10cm is the minimum distance between two mobile phones, showing that data was successfully sent to other devices. However, they found that the transmission rate is low, limited to 0.09 packets (10 bits per second), since higher rates degrade decoding accuracy [9, 31].

Roy and Choudhury have further proposed Ripple I and II [34, 35]. They have tried implementing AM and FM transmission protocols using a customized vibrator and accelerometer. Results show that they have pushed the transmission rate to 30-200 bits/s and that it is protected against eavesdropping [34, 35]. However, it requires higher-level, self-developed mobile phone modules that ordinary users cannot develop. Finally, Zhao et al. have proposed LeaD [33]. They have modified the OOK encoding scheme to a bit group and decoded the bits using machine learning (ML) models for classification. The best ML model can achieve a transmission rate of 40 bits/s with high accuracy [33].

The above scholars have inspired this project to adopt a Vibration-based scheme for P2P communication. However, previous studies have primarily focused on establishing a basic communication connection and transmitting simple strings for experimental purposes only. They often assume that the channel itself is a trusted and safe one. Implementing a secure protocol for the physical vibration channel remains largely unexplored, particularly in financial applications such as a P2P payment system. The project also aims to bridge this gap and demonstrate the feasibility and potential of the security-vibration-based P2P payment system.

## **2.3 Cryptography Framework for Vibration Channels**

### **2.3.1 Security for P2P Mobile Payment Systems**

P2P Mobile payment system security is essential for the users and service providers. Transaction data must be protected when used. The desired security goals in a mobile payment system include integrity, confidentiality, availability, authentication, access control, and non-repudiation [36].

To illustrate further, confidentiality requires that data be restricted to the parties involved in the transaction process and protected against passive attacks [37, 38]. Integrity ensures the system's accuracy during data processing, ensuring that data are not modified in a complete and timely manner [37, 38]. Moreover, non-repudiation ensures that it is a specific user's transfer transaction messages and prevents them from denying the services [38, 39]. Availability ensures the mobile payment system is accessible upon the user's request [38, 39]. Next, authentication requires verifying the users and the data origins [38, 39]. Finally, access control ensures that only authorized personnel can access the mobile payment system and blocks unauthorized personnel from accessing it [38, 39]. This highlights the significance of secure P2P mobile payments and the cryptographic framework, including symmetric and Asymmetric encryption, PKI, and digital signatures. These provide foundational tools to safeguard the mobile payment system and its security goals.

However, cryptographic challenges will arise in a low-bandwidth environment due to constraints, such as a vibration channel. The subsection below discusses the relevant issues.

### **2.3.2 Traditional Cryptographic Challenges in Low-Bandwidth Environments**

As mentioned in section 2.2.2, the vibration-based P2P communication only has a 10-200 bits/s transmission rate, which is significantly slower compared to other P2P communications used in mobile payment systems like Bluetooth with 1-2Mbps (Mentioned in 2.2.1) and NFC with up to 424Kbps [40]. Moreover, Mobile devices often operate under constraints such as limited processing power, memory, and battery capacity [41]. The twofold challenge has created a low-bandwidth, resource-constrained environment for the P2P payment system using vibration.

However, using high-overhead traditional cryptographic protocols is unsuitable for a vibration P2P payment system, as it is difficult to process both vibration communication and cryptography in parallel. For example, previous research indicated that asymmetric encryption and PKI in vibration-based channel devices result in resource drain due to computational complexity and significant communication overhead on low-bit-rate channels [11]. Moreover, running a conventional cryptographic algorithm on a mobile platform consumes significant resources, such as CPU time and memory, leading to rapid battery drain and slower transactions [42, 43].

Yet cryptography is necessary to establish a secure mobile payment system. It must not be omitted to achieve the security objectives. Therefore, Lightweight Cryptographic (LWC) algorithms are

essential in mobile payment systems to balance security and performance. The LWC framework provides strong security properties while minimizing the computational power to ensure a user experience and transaction time.

### **2.3.3 Effective Lightweight Cryptographic (LWC) Framework**

The LWC Framework is not a new concept in cryptography. DES Lightweight extension (DESL) was first introduced in 2007 as a lightweight version of DES encryption, and the idea is simple: take DES and lighten it so that mobile devices can perform encryption at a low cost [44]. All LWCs share common goals: run light and meet security goals, no matter how the algorithm is implemented. As Johnny pointed out, the LWC is feasible on mobile devices, and many lightweight API-based cryptography libraries are supported in the mobile application environment [45]. At the same time, many LWC frameworks are supported. Our studies will focus on reviewing the algorithm used in our project.

Previous research has shown that symmetric cryptography is one of the lightweight solutions for the vibration communication channel [11]. Symmetric algorithms are generally faster and require less computational power than asymmetric algorithms, making them ideal for devices and communication channels with limited processing power, such as smartphone motion sensors [46]. The Advanced Encryption Standard (AES) is a symmetric LWC that provides confidentiality and authentication. AES is a block cipher that uses fixed-length keys and block sizes for encryption and decryption, and provides strong security guarantees with key sizes of 128 to 256 bits. Moreover, AES is included in many international and regional standards, making it a default option for secure communication in constrained environments [46]. From a lightweight perspective, many smartphones currently include an AES hardware module that enables encryption without burdening the CPU or memory. In symmetric cryptography, increasing key sizes does not significantly increase energy consumption, as the key expansion step in AES accounts for only a small fraction of the total computation [46, 47]. The results concluded that AES may be suitable for the vibration communication channel, such as the LWC.

However, we must consider the constraints of the AES encryption approach when implementing it in the studies. First, the data block size will be a challenge. As mentioned in the literature, the key size is fixed relative to the block size; if we choose a larger key size for better security, the block size will increase accordingly. However, 128-bit blocks can be excessive for devices that only transmit small messages [46]. We must consider the tradeoff between security, computation



power, and time when designing an AES encryption scheme for the vibration channel. The solution lies not in changing AES, but in changing the operation mode, as the operation's use affects how smaller or larger messages are handled within a single block. In traditional protocols like AES-Cipher Block Chaining (CBC), when the message size is smaller than the fixed block size, padding is triggered and extra data is added to fill the blocks [48]. The method is not ideal for a vibration communication channel, as the limited bandwidth is wasted processing the redundant data introduced by padding. One mode that is better suited to a lightweight mindset is AES-CTR (AES-Counter). The approach converts the block cipher into a stream cipher, which uses a counter (or nonce) and a key to generate a key stream, which is XORed with the plaintext to form the ciphertext for communication (*Figure 2.2* below shows a typical flow of CTR mode) [49, 50].

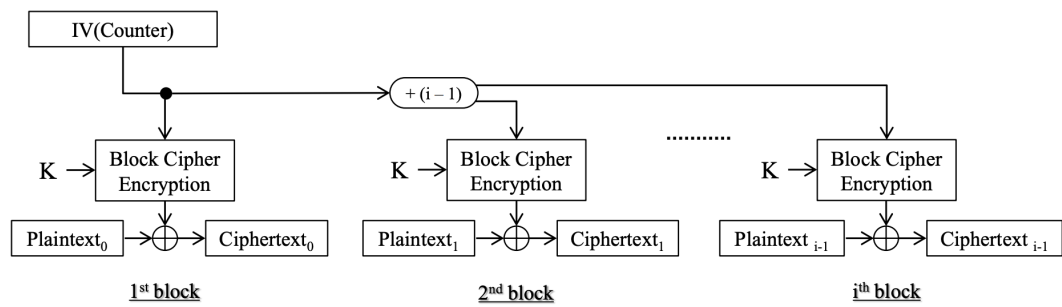


Figure 2.2: CTR Mode Encryption [50]

The CTR encryption shows that it is more parallelizable and doesn't require padding of the block while creating a ciphertext the same size as the small message being sent. It offers higher flexibility in determining the message size, which may suit a lightweight approach.

Moreover, Prior studies indicate that symmetric cryptography, which relies on exchanging a shared secret key between two devices, presents challenges in the vibration channel [11]. *Figure 2.3* shows that symmetric key cryptography uses the same private key for both encryption and decryption, and that both parties require a pre-shared secret key to communicate securely [51].

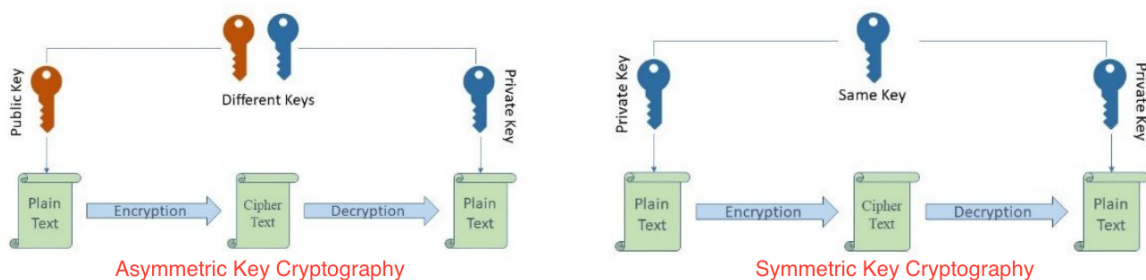


Figure 2.3: Asymmetric & Symmetric Key Cryptography processes comparison [51]

This presented the challenge of setting up and exchanging the key before communication in a lightweight manner, as the energy consumption and resource demands of AES depend not only on the process but also on the key setup cost [52].

A form of data transmission protection is password-based encryption, which uses a password or passphrase entered by end users as a cryptographic key [53]. It is an ideal lightweight key establishment protocol because it relies on a few message exchanges and simple cryptographic operations, making it suitable for a low-bandwidth channel [54]. Yet, it is very vulnerable from a security perspective since users will prefer weak passwords that are easy to remember; these passwords are generally short and easy to guess, like using their birthday as a password, which is highly susceptible to dictionary attacks that attackers try all possible passwords, until they find the correct one from a defined file [55, 56]. To mitigate password weaknesses, the researcher introduced the “Password-Based Key Derivation Function version 2” (PBKDF2) to strengthen passwords before use, thereby establishing a robust secret key. The PBKDF2 is denoted by:

$$DK = PBKDF2(p, s, c, dkLen).$$

The algorithm will finally output a derived key  $DK$  where  $p$  is the passphrase input by the user, a random salt  $s$ , an iterated loop counter  $c$ , and the secret key length  $dkLen$  [57]. The PBKDF2 added salt to prevent rainbow table attacks and applied hashing like HMAC-SHA256 many times on a defined iteration loop that establishes the secret key for the AES-secured. It is also a suitable approach in a lightweight context since it is simple to implement. The key length and iteration count can be tuned based on mobile device capabilities, which provide considerable flexibility for customization [58, 59]. Password-based encryption with PBKDF2 is a suitable lightweight method for key setup and exchange in our project.

Lastly, we should consider a cryptographic protocol to protect authentication and integrity during message transfer. Message Authentication Code (MAC) is a symmetric cryptographic mechanism that allows two users with the same shared key to authenticate the data transmitted. *Figure 2.4* illustrates the whole simple process of the MAC protocol. First, the sender computes a MAC tag using the message  $M$  and key  $K$ ; then it sends the message with the tag as a block to the destination. Next, the receiver computes a new MAC tag using  $M$  and  $K$ , and compares it with the received MAC tag. If it is equal, the receiver can conclude it is not altered during transmission [60].

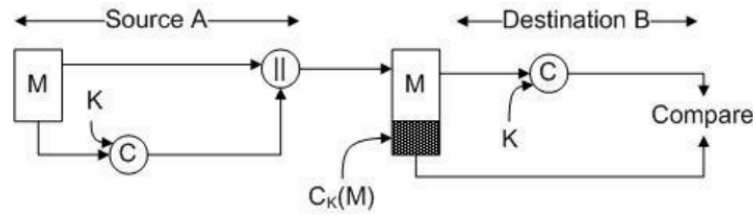


Figure 2.4: Overview of the Message Authentication Code [60]

From a security perspective, if we assume only the two parties know the secret key and the calculated MAC tag matches the received MAC tag, then:

- Integrity: The tag helps verify that the message has not been modified, as any alteration would cause verification to fail [60, 61].
- Authentication: The tag assures the message is from an authentic sender, proving that the data originated from a user who knows the secret key [60, 61].

The studies have proven that we should consider the MAC design when implementing a cryptographic framework. Nevertheless, given the limited processing power of the vibration-based communication channel. We should also consider using a lightweight approach to the MAC. Previous studies have proposed a lightweight MAC using a “Hash-based MAC” (HMAC) approach to ensure security goals while keeping computation costs low in low-bandwidth channels, such as those found in IoT devices [62]. The simple structure of the HMAC is relatively clean, where the function is  $t = H(K, M)$ , where  $t$  is the MAC tag of the Message  $M$  and the key  $K$ , and a hash function  $H$  using a compression algorithm like SHA-256, and the complexity of the whole function depends on how the key and message are constructed [63]. Consider the designs of lightweight HMACs proposed by previous scholars, which use only hashing, the XOR operation, and data concatenation. These are computationally inexpensive, reduce communication overhead by minimizing the number of bits transferred, and maintain the original security measures [62, 64].

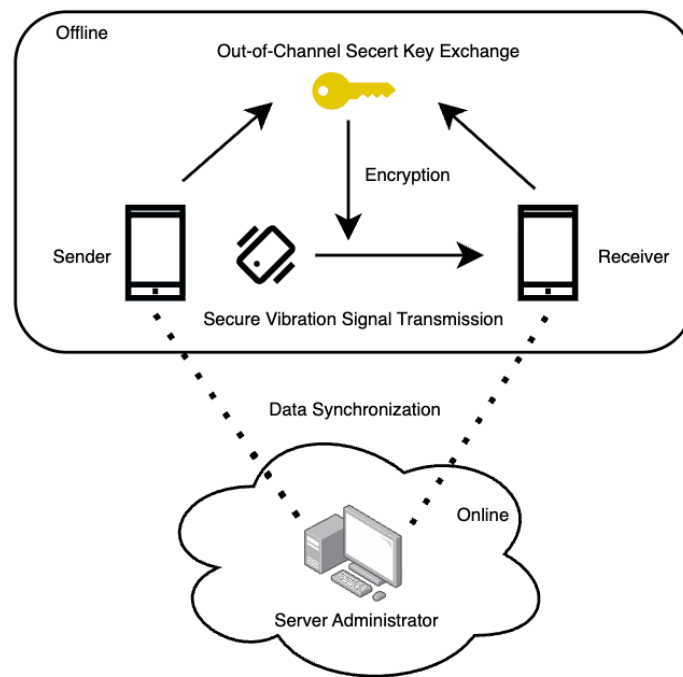
To conclude, Section 2.3.3 provides clear insight into LWC components for securing communication over constrained channels such as vibration. Previous research supports the use of AES, out-of-band password-based encryption with PBKDF2, and HMAC with a lightweight framework, which help fulfill the security goals of a P2P payment system. However, while these protocols are well-researched individually, there is a significant gap in integrating protocols within

the vibration channel P2P payment mobile system. The project hopes to design and evaluate a lightweight cryptography framework for the secure P2P vibration payment system.

### 3. Proposed Design

This section provides an overview of the design for vibration-based communication in the P2P payment system, along with the software engineering (SE) approach considered. Next, it will discuss how the system will be evaluated in general, including the testing procedures and setup. Lastly, the project's development cycle will be introduced.

#### 3.1 Overall P2P Vibration Mobile Payment System



*Figure 3.1: System diagram of the proposed solution*

*Figure 3.1* illustrates the overall system architecture of the vibration-based P2P payment system. The system will consist of three major components. The first components focus on the mobile payment system's functionalities, including user-facing features and the design of vibration communication. The second component focuses on a payment management system for administrators, including transaction logging and user data synchronization. The last component focuses on the lightweight cryptographic design, incorporating secret-key exchange and the encryption scheme for vibration communication. All transactions can be completed in a physical, offline layer, except when data synchronization and transaction log saving are required on the server side.

Below are the overall technical framework and libraries used for this project:

- **Coding environment:** We will use Android Studio IDE and Kotlin to develop each component of the project.
- **System's UI and functionalities:** The project will use Jetpack Compose, an open-source toolkit for building native UI on Android.
- **Vibration communication channel:** The Android Haptic APIs will be used to generate and control the vibration channel, and the Android Sensor Framework (*TYPE\_ACCELEROMETER* & *SensorEventListener*) will be utilized to receive and listen to the vibration data.
- **Lightweight Cryptographic Protocol:** The project will apply javax.crypto/security API for cryptographic operations, including SecretKeyFactory, Mac, and SecureRandom.
- **Database:** A local database file will be used to store user data and transaction records, serving as a test database and backend for various applications, including authentication, storage, and database management.

### 3.2 Functionalities of the Proposed System

Users can log in, log out, and create an account for the payment system. Moreover, they can also update their status, including their displayed username and passwords. This provides users with an authentication and customization function.

Moreover, users can check their payment history on the history screen, which is categorized into “Pay” and “Receive payment” records. All payment logs will be sorted by date and time in descending order by default for improved display. Users can also search for specific transaction records by using keywords related to the transaction details.

Furthermore, users can send and receive payments from other users. If users choose “Send Payment”, the system's vibration encoder will encode the payment's vibration pattern using the customized OOK algorithm and send it to the receiver for payment. If users choose “Receive Payment”, the vibration decoder in the system will listen to the vibration pattern from the transmitter and decode the payment vibration pattern using the adaptive threshold algorithm, and finally confirm the payment.

Before encoding or decoding the vibration payment pattern, the system requires both the transmitter and the receiver to enter a verbally agreed-upon secret passphrase. The system then

generates a Pre-Shared Secret (PSS) key by processing this passphrase, which is used to encrypt and decrypt the vibration payment data.

In Section 4, this paper will introduce the detailed implementation of the vibration communication channel and the lightweight cryptographic protocol.

Lastly, if users are online (with an internet connection), they can update their account status and transaction log on the online server. From the server administrator's management side, they can check the transaction log and user status on the server.

Figure 3.2 below illustrates a use case diagram that describes the system's high-level functions and identifies the interactions between the system functions and its users.

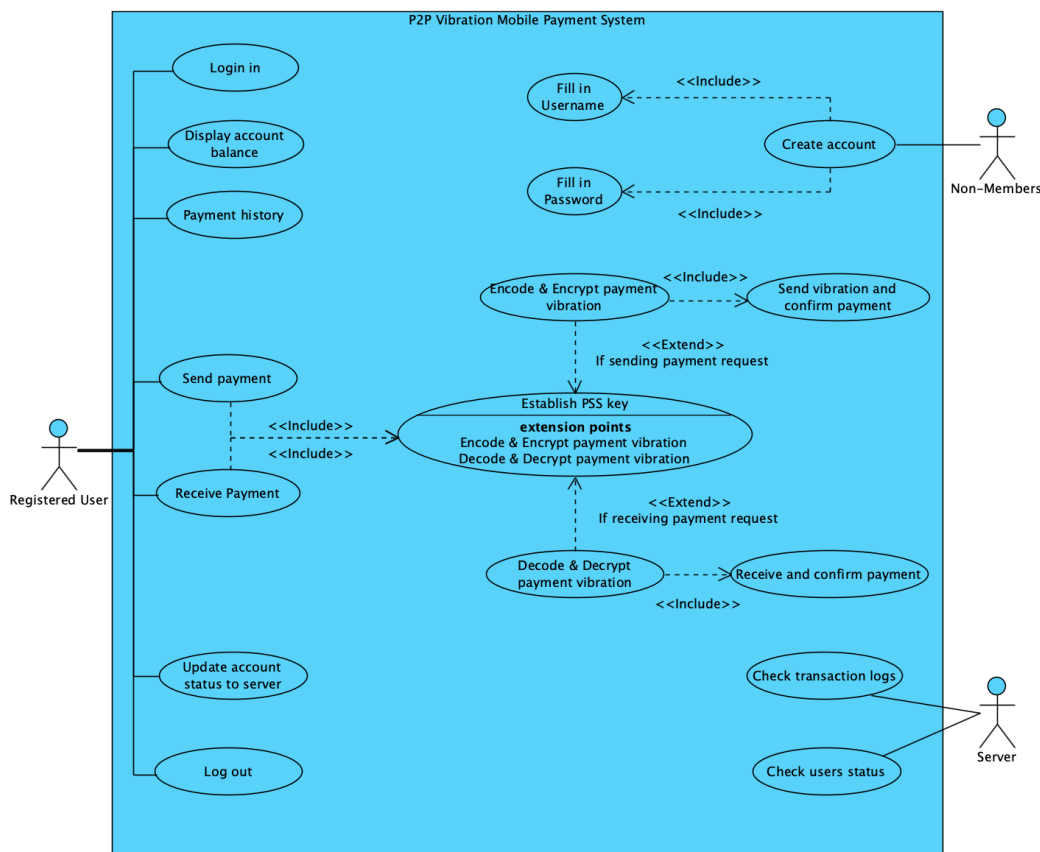


Figure 3.2: Use case diagram of the proposed solution

### 3.3 Software Engineering Approach of the Proposed System

We should also consider a software engineering (SE) approach, such as design principles and patterns, before implementing the whole system.

First, we should consider the design principles. The Single Responsibility Principle (SRP) ensures that every class or object has only one job, thereby increasing maintainability by making components highly reusable and easy to test. For example, vibration design should ensure that the vibration encoder is only responsible for encoding the payment request to a vibration pattern and the decoder should only perform the reconstruction of the original data of the vibration.

Furthermore, the Open-Closed Principle (OCP) should be considered: the system's functions should be open to extension and closed to modification. OCP helps minimize the impact on existing code when making changes and makes it easy to add new functions. Abstraction and subclassing techniques can help to achieve the principle. For example, the Lightweight Cryptographic Protocol always requires encryption and decryption of the message. Their used attribute and behavior will always remain the same, and the only difference is the cryptographic algorithm used in each process. Therefore, the OCP can ensure that the class remains open to extending the protocol's use while maintaining the behavior and attributes defined in the design.

Lastly, this project should also consider the KISS principle, which emphasizes keeping the project simple and reliable to solve complex problems. For example, in this project, we are evaluating the secure vibration channel but not developing a complex mobile payment system, so we should implement a simple function to demonstrate each use case using the proposed solution.

Next, we should consider the design pattern. The singleton pattern should be considered; this pattern ensures that only one instance of a class exists in the entire application. It provides better control as there is only one vibration controller for physical vibration, and prevents duplicated parts from performing the same tasks.

Additionally, the observer pattern is considered, where there is a listener is notified of changes from the publisher. In our project, the UI of each component should observe changes to the variable and automatically update its display information. For example, the home menu monitors the user's balance and displays the user's current balance.

Lastly, to ensure a program performs tasks in the background without halting its main execution, it is necessary to prevent the vibration channel from requiring the user to idle the mobile phone for sending and receiving vibration patterns, as this would completely freeze the application's interaction. Kotlin Coroutines are used for structured concurrency and thread confinement. For



example, the send payment coroutine in HomeMenu runs in the background while concurrently updating the transaction status to prevent memory crashes or loss.

Figure 3.3 illustrates a simple class diagram of the proposed design, which incorporates the considered design patterns and principles.

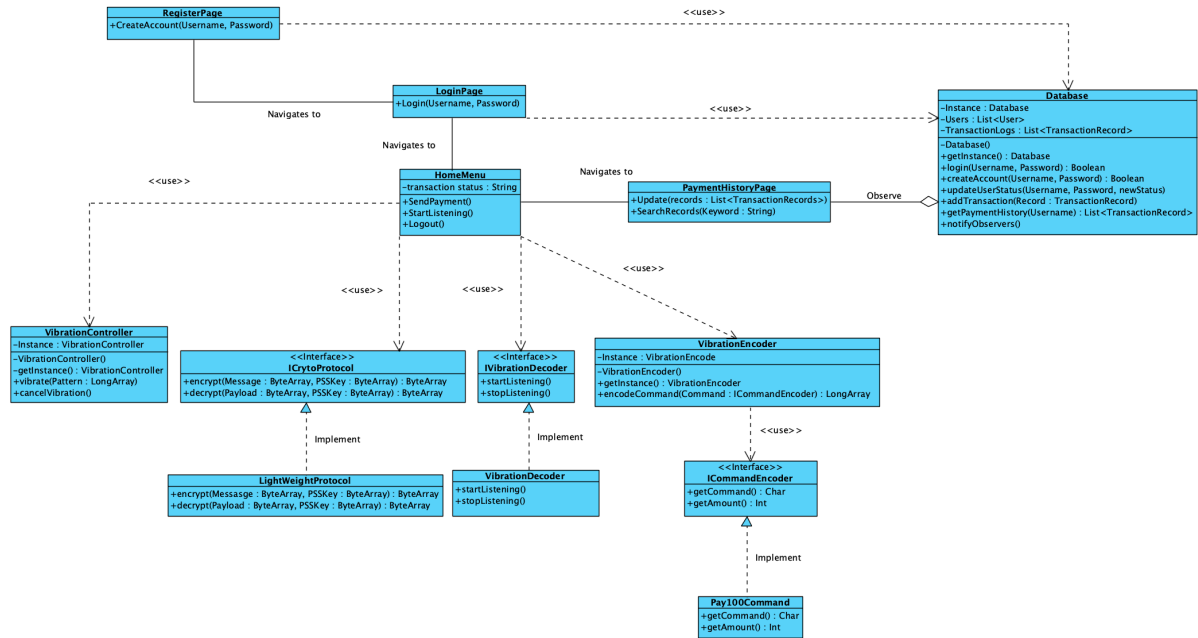


Figure 3.3: Class diagram of the proposed solution

### 3.4 Performance Evaluation Design

The final prototype will be tested on two Android phones. A Bottom-up approach will be used to test the system's functionality, followed by performance and penetration tests for testing the vibration algorithm and security protocols.

#### Unit Testing

During system development, we will perform unit tests for each developed class, following the functions specified in Section 3.2.

#### Integration Testing

After completing each phase, we will integrate the subset of each class into a subsystem, such as a combination of the UI and the vibration transaction. The purpose is to ensure that the subsystem's functions interact properly.

Both unit and integration testing will be performed using White Box testing Approach, which helps to detect bugs during implementation.

### **System Testing**

In the final phase of implementation, we will integrate all subsystems into a comprehensive P2P vibration-based mobile payment system to test its functionality without faults in a controlled environment. It is a Black-Box Testing Approach to evaluate the system against defined use-case scenarios and to demonstrate usability results.

### **Performance Testing**

We will test the accuracy of transaction transmission using the designed vibration-encode/decode algorithm under different scenarios. It helps validate and strengthen the design's viability and quantify its reliability for further studies on the vibration channel in the mobile payment system.

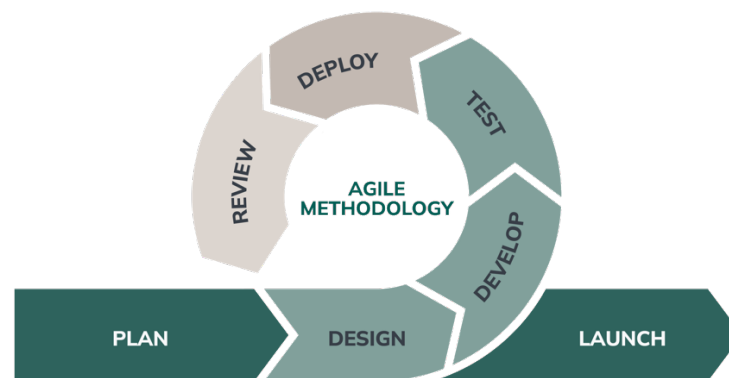
### **Penetration Testing**

Penetration testing (Pen test) is a simulated cyber attack designed to exploit vulnerabilities in a system before it is publicly released. It helps to validate if the theoretical cryptographic design is implemented securely in practice and achieves the security goals.

The testing results and discussion will be presented in Section 5.

## **3.5 Software Development Methodology and Stages**

The project follows an Agile software development approach, with iterative development cycles. (see *figure 3.4*). Consider that the project has only a 6-month development period, from planning to application launch. The proposed SE approach is suitable for a rapid, incremental, and continuous delivery in a short-term period.



*Figure 3.4: Diagram of the Agile software development cycle*

The standard for this practice is that, after initial planning and goal, and use-case definition, we will start by designing a simple software application with limited functions. Next, we will begin developing another function and integrate it into the system by looping it several times, adding each function one at a time. Every time we add a new function, we will review and test its behaviour to ensure it is well-matched with the proposed system.

Therefore, the implementation has been distributed into five phases (1-5) to allow greater flexibility. The schedule is outlined below (More details about the high-level project schedule plan and Gantt Chart can be found in Appendix B):

- a) Phase 1: P2P Payment System Interface (September 2025)
- b) Phase 2: Basic Vibration channel development (October – November 2025)
- c) Phase 3: Cryptographic framework development & Integration (December 2025 – January 2026)
- d) Phase 4: System Integration & Testing (February 2026)
- e) Phase 5: Final Preparation & Documentation (March 2026)

*Figure 3.5* illustrates the agile development cycle for our project. We developed our system by focusing on small components in each iteration, integrating them one by one in each phase. Eventually, this approach will lead to the creation of our secure P2P payment system through vibration.

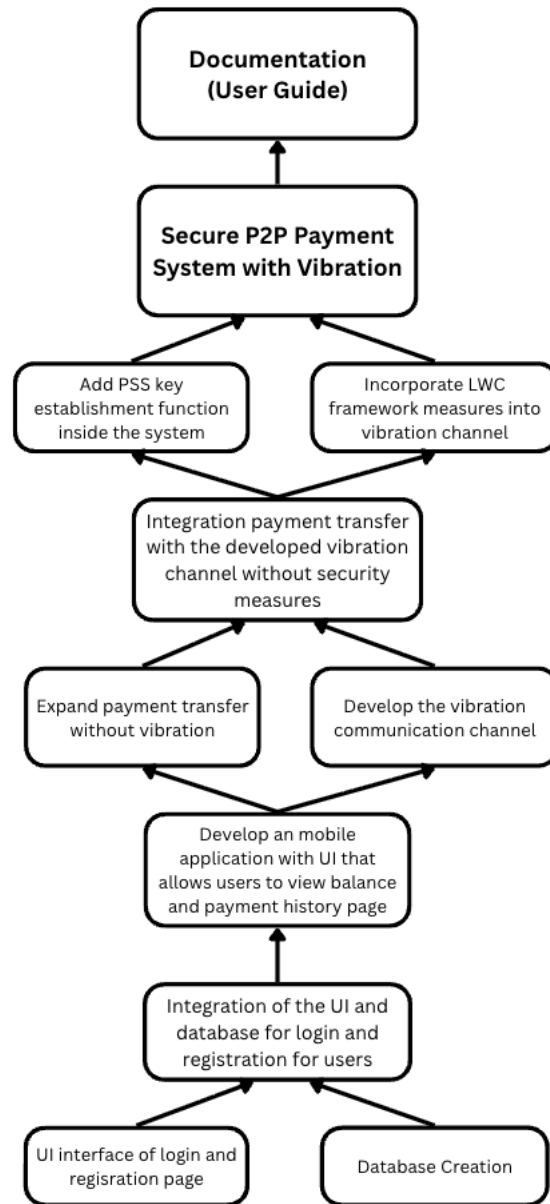


Figure 3.5: Agile software development cycle for the proposed system

## 4. Detailed Methodology and Implementation

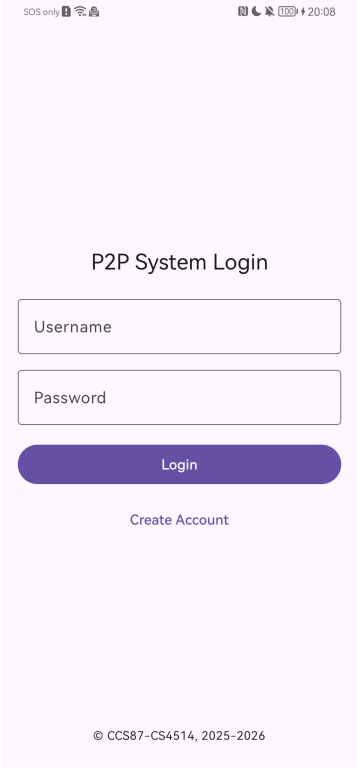
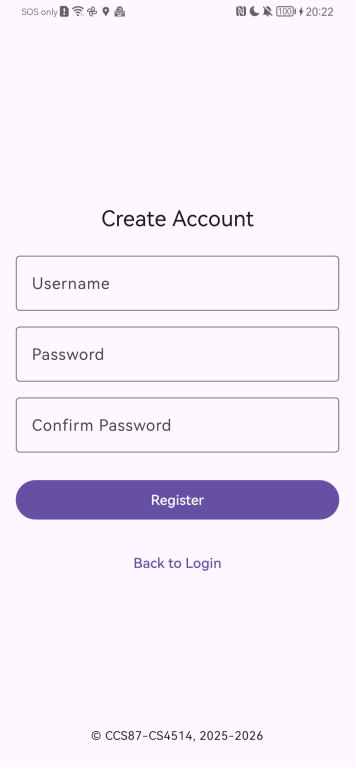
This section provides a detailed implementation of the system's components, including its overall interfaces & database, the design of the vibration communication channel, and the Lightweight Cryptographic Protocol.

### 4.1 User Interface Design

Based on the defined functionalities in Section 3.2, each function was implemented on different pages inside a mobile application. The interface design is intended to be simple and easy to navigate, with improved usability. As it is also a proof-of-concept application for the secure vibration channel, the UI design won't be the main focus of development. The P2P Vibration Mobile Payment System comprises five main pages: Login, Home, History, Transaction, and Management.

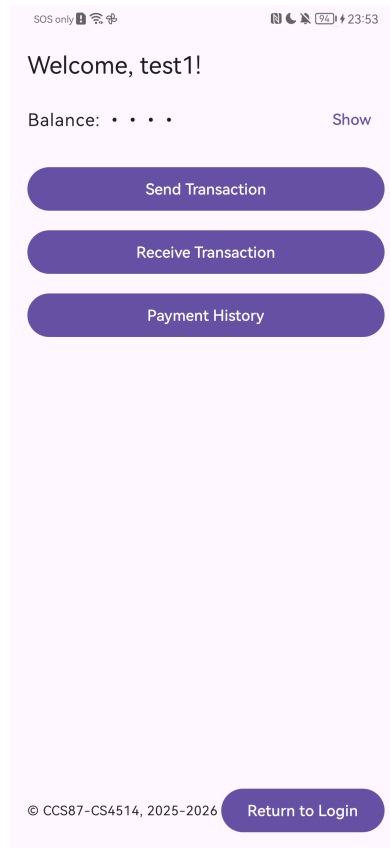
#### Login Page

This page allows users to log in to the payment system if they have an existing account in the database. They can create an account on the registration page if they are new users.

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |
| <i>Figure 4.1: Layout of the login page</i>                                         | <i>Figure 4.2: Layout of the Registration page</i>                                   |

## **Home Page**

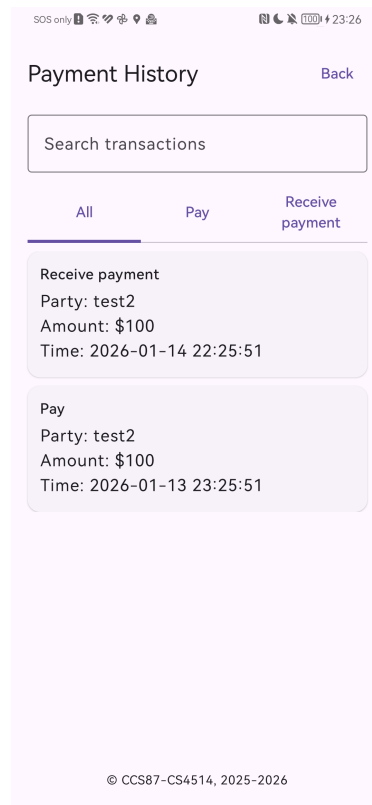
This is the main directory page, which contains all the functions that users can perform, allowing them to navigate to other pages, such as sending transactions or checking their payment history (See *Figure 4.3*).



*Figure 4.3: Layout of the User Home menu page*

## **Payment History Page**

On this page, users can view their transaction history, including payment and receipt records with other users. They can also search for the required transaction records by entering keywords in the search bar (See *Figure 4.4* below).



*Figure 4.4: Layout of the payment history page*

### **Transaction Page**

The users can choose to transact \$100 (“a”) or \$200 (“b”), or receive payment from another user. Moreover, they must set up the PSS Key for encryption before the transaction. Detailed interaction will be shown in the figures below.

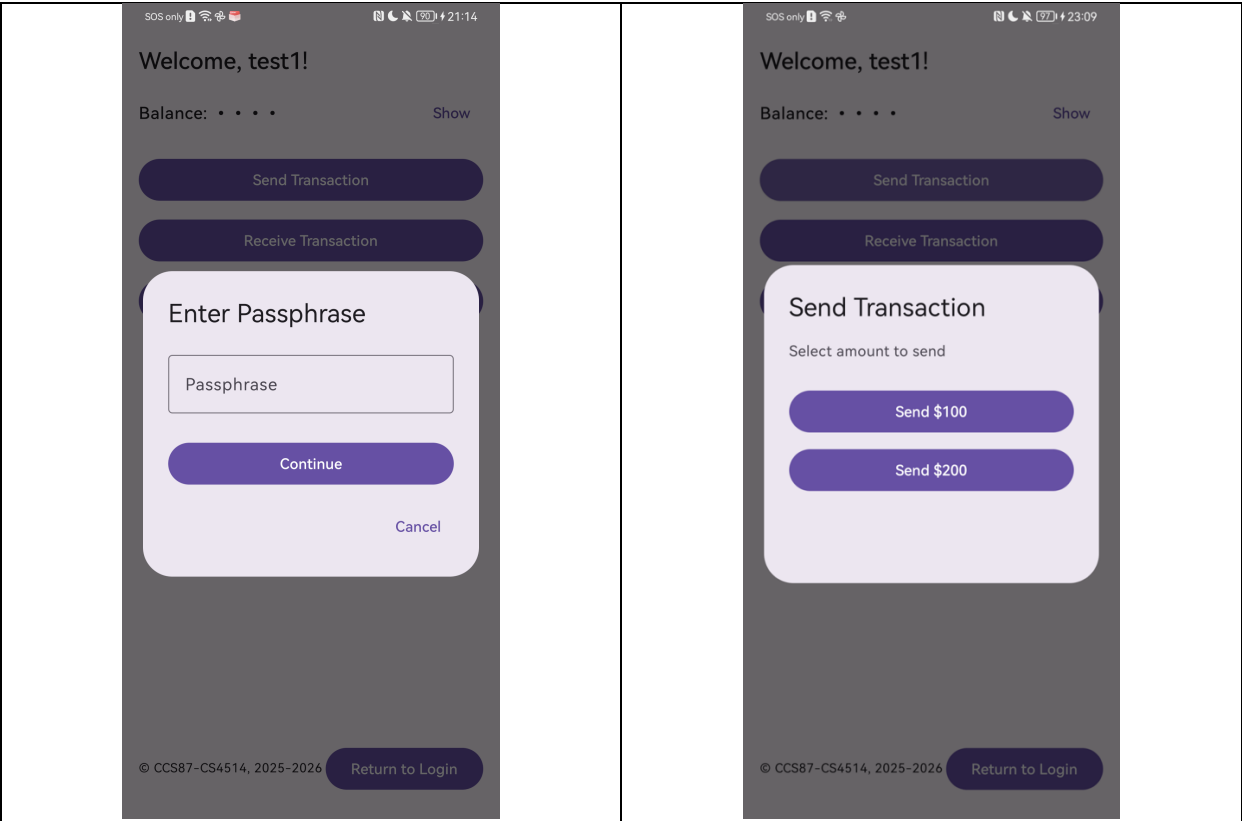


Figure 4.5: Passphrase input for PSS Key set up

Figure 4.6: User can choose the payment amount

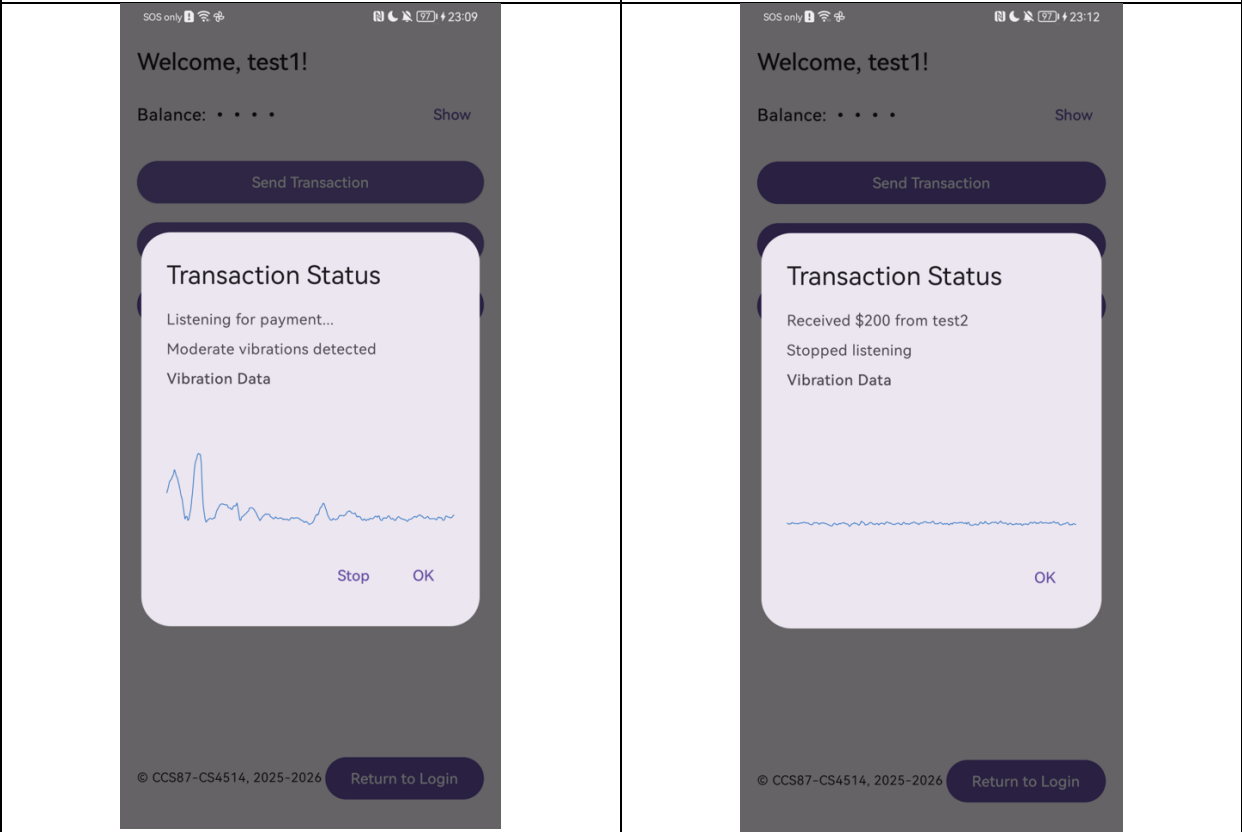


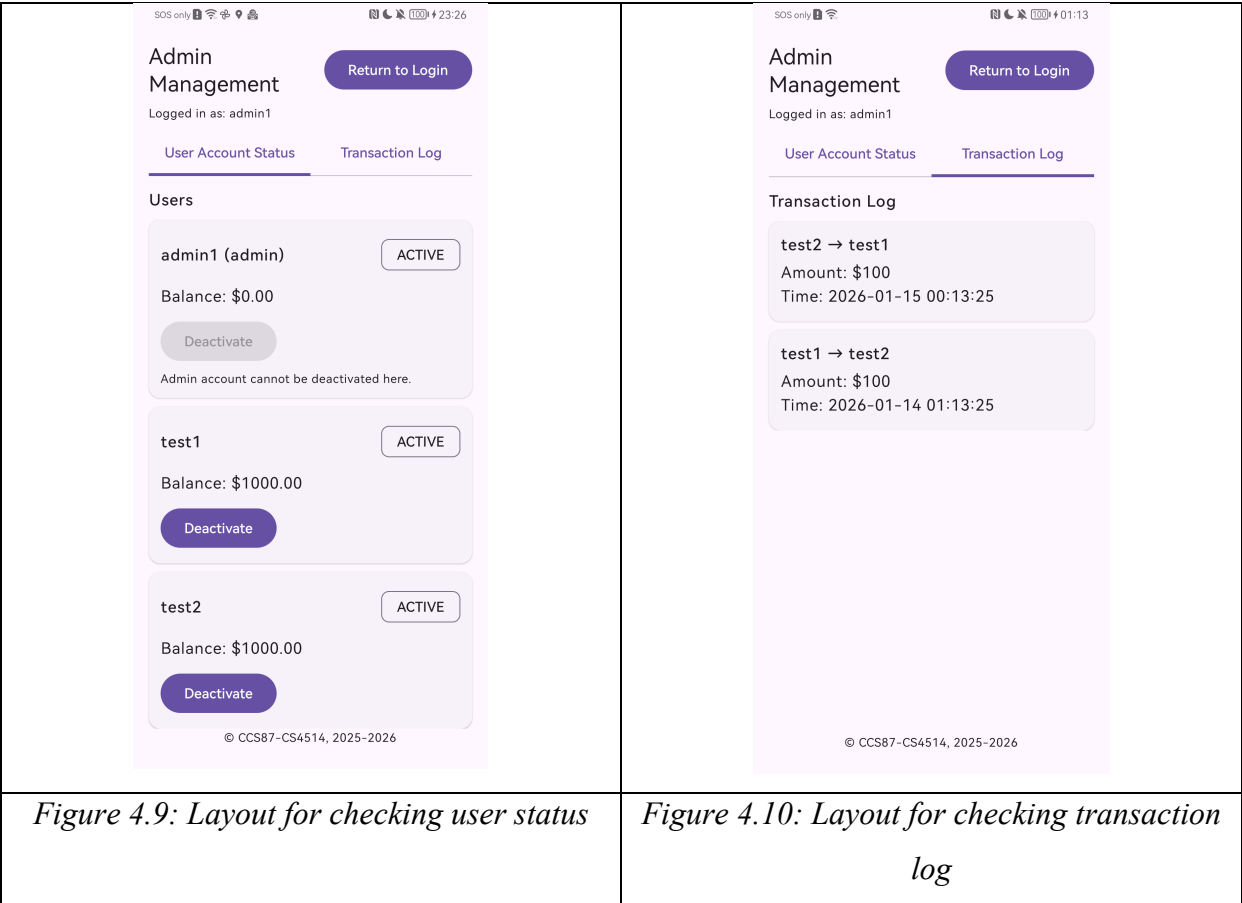
Figure 4.7: Transaction status – Vibration detection

Figure 4.8: Completion transaction status



**Admin Management Page**

On this page, administrators can view the remaining balances of all users and manage their account statuses. Moreover, they can review all the transaction records users have previously performed.



## 4.2 Database Design

In our proposed P2P payment system, a simple database was implemented. As it is a proof-of-concept application for the secure vibration channel, the database also won't be the main focus of development. The ER diagram of the database is shown below (PK = Primary key; FK = Foreign Key).

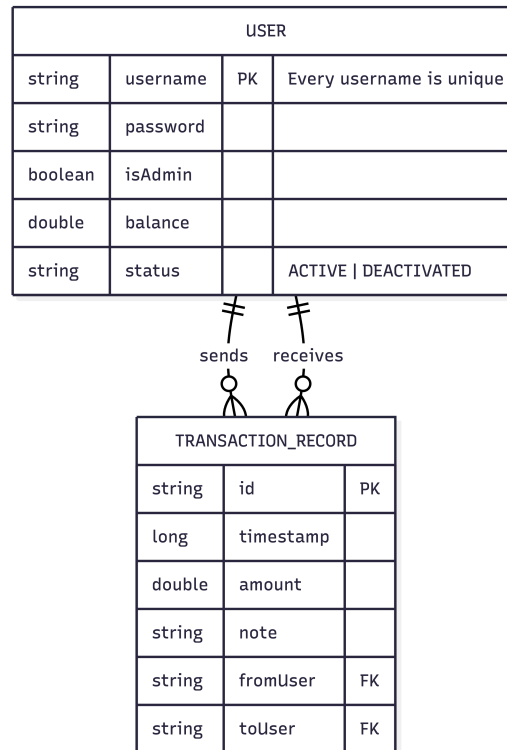


Figure 4.11: ER diagram of the database

## 4.3 Vibration Communication Channel Design

An overview of the design of vibration communication will be presented below. It is separated into two major components: 1) Vibration Encoder and 2) Vibration Decoder. The minor components that support the encoder and decoder: the Vibration Controller and Accelerometer Monitor, which use the built-in vibrator and accelerometer in the smartphones. *Figure 4.12* below shows the overall design of the vibration communication channel.

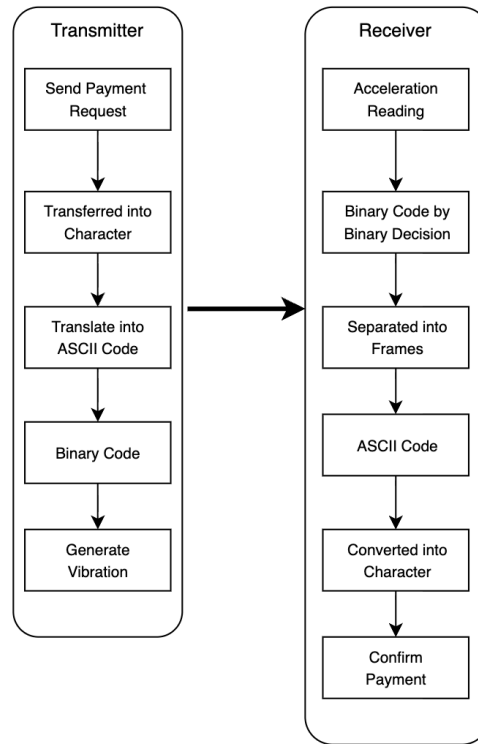


Figure 4.12: An overview of the vibration communication channel

The Vibration Controller is a simple and low-level hardware component that directly interacts with the mobile device's haptic motor. It is responsible for receiving a pre-defined vibration pattern from the encoder and executing it on the device vibrator. Moreover, it also ensures the vibration strength is set to maximum (255) only during vibration and minimum (0) when it is paused.

The Accelerometer Monitor serves as the hardware interface for data collection and conversion. It detects the physical vibration by listening to the device's built-in accelerometer. It collects raw sensor data, stores it in temporary buffers, and then transfers it to the decoder for processing. The details of the two key actions are shown below.

- **Signal conversion:** It collects the raw  $X, Y, Z$ -axis data from the accelerometer and converts it to a single-scale magnitude. The formula is defined as:  $M = \sqrt{X^2 + Y^2 + Z^2}$
- **Data buffering:** It helps transfer all object data into a list and pass it to the Vibration Decoder for analysis.

- **Data Framing:** The encoder first converts the payment request (Data) into characters, such as “a” corresponding to \$100 and “b” corresponding to \$200, and then converts it to a 7-bit ASCII code. Then, the encoder will construct the vibration pattern into a frame format of “Start (00010) + Data (1100001) + End (00011)”, resulting in a total of 17-bit data frame.
- **Modulation:** The data frame will then be modulated using the OOK scheme. Each bit is allocated into a 1-second time slot, and the data rate is fixed at 1 bit per second (bps). Where bit ‘0’ is a 1000ms pause (No vibration), and bit ‘1’ is a 200ms vibration, and padded with 400ms on either side (400ms pause + 200ms vibrate + 400ms pause) to fit the 1000ms window.

---

**Algorithm 1** Vibration encoding algorithm

---

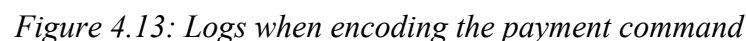
**Require:** Command character  $C$

**Ensure:** Data Frame  $D$

- 1:  $asciiCode \leftarrow ASCII(C)$
- 2:  $binaryData \leftarrow ToBinary(asciiCode, length = 7)$
- 3:  $startMarker \leftarrow "00010"$
- 4:  $endMarker \leftarrow "00011"$
- 5:  $fullBinary \leftarrow startMarker \parallel binaryString \parallel endMarker$
- 6:  $D \leftarrow \emptyset$
- 7: **for all**  $bit$  **in**  $fullBinary$  **do**
- 8:   **if**  $bit == '1'$  **then**
- 9:     Append 400ms to  $D$
- 10:    Append 200ms to  $D$
- 11:    Append 400ms to  $D$
- 12:   **else**
- 13:     Append 1000ms to  $D$
- 14:   **end if**
- 15: **end for**
- 16: **return**  $D$

---

The results are a 17-second vibration sequence, such as [1000, 1000, 1000, 400, 200, 400, ...], which is then passed to the Vibration Controller for execution. *Figure 4.13* illustrates the process that occurs after users select the payment amount.



The Vibration Decoder is a primary component for the Receiver. It analyzes the raw sensor data collected by the Accelerometer Monitor to reconstruct the original data. This process is triggered automatically after a fixed buffering period and involves two main stages.

- **Bit Reconstruction:** A function is available for converting analog accelerometer data back into a digital bitstream. It iterates through the buffered list (Depending on the size of the transaction) in a discrete 1-second window corresponding to the synchronous bit rate. For each window, it checks whether the calculated magnitude exceeds a set threshold (required magnitude) a sufficient number of times. If vibration is detected, a “1” is appended to the binary string and vice versa. For the threshold and required count, these values can be adjusted to any desired level based on specific needs.
- **Pattern Matching:** The constructed binary string will be passed to another function that searches for the 17-bit data frame to verify its validity. This is achieved by finding the start marker [i, ..., i+5] and subsequently checking the 12 bits for the end marker [i+12, ... i+17]. The data frame in the middle is then extracted for processing.

Table 4.2 shows a simple illustration of the decoding algorithm:

---

**Algorithm 2** Vibration Decoding Algorithm

---

**Require:** List of Accelerometer datas  $A$ , Threshold  $\tau = 9.65$   
**Ensure:** Decoded Commands  $C$

**Stage 1: Bit Reconstruction**

```

2:  $Bits \leftarrow \emptyset$ 
    $T_{bit} \leftarrow 1000ms$ 
4: for each time window  $W$  of duration  $T_{bit}$  in  $A$  do
    $count \leftarrow$  Number of samples in  $W$  where value  $> \tau$ 
    $maxVal \leftarrow$  Maximum sample value in  $W$ 
   if  $count > 5$  AND  $maxVal > \tau$  then
8:   Append '1' to  $Bits$ 
   else
10:  Append '0' to  $Bits$ 
   end if
12: if Length of  $Bits \geq 17$  then
   break
14: end if
end for

16: Stage 2: Frame Matching
    $StartMarker \leftarrow "00010"$ 
18:  $EndMarker \leftarrow "00011"$ 
    $C \leftarrow \emptyset$ 
20: for  $i \leftarrow 0$  to Length( $Bits$ ) do
    $window \leftarrow Bits[i \dots i + 17]$ 
22: if  $window$  starts with  $StartMarker$  AND ends with  $EndMarker$  then
    $DataFrame \leftarrow window[5 \dots 11]$ 
24:  $AsciiVal \leftarrow BinaryToDecimal(DataFrame)$ 
    $Command \leftarrow Char(AsciiVal)$ 
26: if  $Command \in \{'a', 'b'\}$  then
   Add  $Command$  to  $C$ 
28: end if
   end if
30: end for
   return  $C$ 

```

---

Table 4.2: Pseudocode of the decoding algorithm

Finally, if only a single valid pattern is found, the decoder converts the ASCII value back and identifies it as the payment command to process the required transaction. If multiple valid patterns are detected due to signal noise, the user is presented with a list of possibilities to help them make the final selection. The figures below illustrate the process that occurs behind the decode algorithm during receiver listening to the vibration patterns.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> Data: 8400 points, Max: 9.73, Vibrations: 0 *** HIGH VIBRATION: 9.70 at 16812ms *** *** HIGH VIBRATION: 9.68 at 16822ms *** *** HIGH VIBRATION: 9.67 at 16836ms *** *** HIGH VIBRATION: 9.69 at 16845ms *** *** HIGH VIBRATION: 9.68 at 16856ms *** Data: 8500 points, Max: 9.70, Vibrations: 0 Data: 8600 points, Max: 9.61, Vibrations: 0 Data: 8700 points, Max: 9.61, Vibrations: 0 Data: 8800 points, Max: 9.61, Vibrations: 0 Data: 8900 points, Max: 9.61, Vibrations: 0 Data: 9000 points, Max: 9.61, Vibrations: 0 </pre> | <pre> BIT DETECTION: Duration=20113ms, Expected bits=20 Bit 0: max=9.63, avg=9.60, vibrations=0 -&gt; 0 Bit 1: max=9.62, avg=9.60, vibrations=0 -&gt; 0 Bit 2: max=9.63, avg=9.60, vibrations=0 -&gt; 0 Bit 3: max=9.74, avg=9.60, vibrations=14 -&gt; 1 Bit 4: max=9.62, avg=9.59, vibrations=0 -&gt; 0 Bit 5: max=9.71, avg=9.59, vibrations=22 -&gt; 1 Bit 6: max=9.71, avg=9.59, vibrations=21 -&gt; 1 Bit 7: max=9.61, avg=9.59, vibrations=0 -&gt; 0 Bit 8: max=9.61, avg=9.59, vibrations=0 -&gt; 0 Bit 9: max=9.61, avg=9.59, vibrations=0 -&gt; 0 Bit 10: max=9.61, avg=9.59, vibrations=0 -&gt; 0 Bit 11: max=9.75, avg=9.59, vibrations=25 -&gt; 1 Bit 12: max=9.60, avg=9.59, vibrations=0 -&gt; 0 Bit 13: max=9.61, avg=9.59, vibrations=0 -&gt; 0 Bit 14: max=9.61, avg=9.59, vibrations=0 -&gt; 0 Bit 15: max=9.74, avg=9.59, vibrations=26 -&gt; 1 Bit 16: max=9.73, avg=9.59, vibrations=20 -&gt; 1 COMPLETED BIT DETECTION: 17 bits RAW BINARY DETECTED: 000101100000100011 (17 bits) FINAL DECODE: Searching for pattern in 17 bits PATTERN FOUND at position 0: start=00010, data=1100001, end=00011 VALID COMMAND: 'a' (binary: 1100001, ASCII: 97) SUCCESS: Single command 'a' decoded </pre> |
| <p><i>Figure 4.14: Detection of Vibration during Listening process</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><i>Figure 4.15: Logs when decoding the payment command</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

#### 4.4 Lightweight Cryptographic Protocol Design

The vibration communication channel, as described in Section 4.2, provides a novel method for data transmission. However, the pure data frames are transmitted without any protection, making them may vulnerable to several attacks and failing to meet the security goals of the mobile payment system. This section provides the details of the LWC design to secure the channel. To prevent potential attacks and fulfill the security goals outlined in the literature review (Section 2.3.1), we should implement a simple cryptographic algorithm to secure the communication channel.

As described in the literature review section (Section 2.3.2). Since the vibration channel is low-bandwidth (1 bps), we should consider using the lightweight cryptographic frameworks described in section 2.3.3. Symmetric key cryptography should be used. We should also consider which protocol and how to implement it in a lightweight manner.

Based on the findings of the literature review (Section 2.3.3), the framework is constructed based on three standard lightweight symmetric protocols listed below:

- **Key Derivation:** To solve the key exchange problem for AES symmetric encryption, both the transmitter and receiver will first verbally agree on a shared passphrase. After exchanging the passphrase, the system will process it using PBKDF2 with a static salt and a fixed iteration count. The hard-coded method ensures that both devices generate the same 128-bit PSS key. The PSS key will be stored temporarily and removed after the transaction is complete, and it is not expected to be reused.
- **Encryption Scheme:** Given the transmission time of the vibration channel, we will use AES-CTR. This converts the AES block cipher into a stream cipher, enabling it to generate a keystream of exactly the length required to encrypt the data, thereby avoiding padding that could increase transaction time.
- **Integrity and authentication:** To prevent tampering, a HMAC-SHA256 is used. The 256-bit output is truncated to 16-bit tags and concatenated into the data.

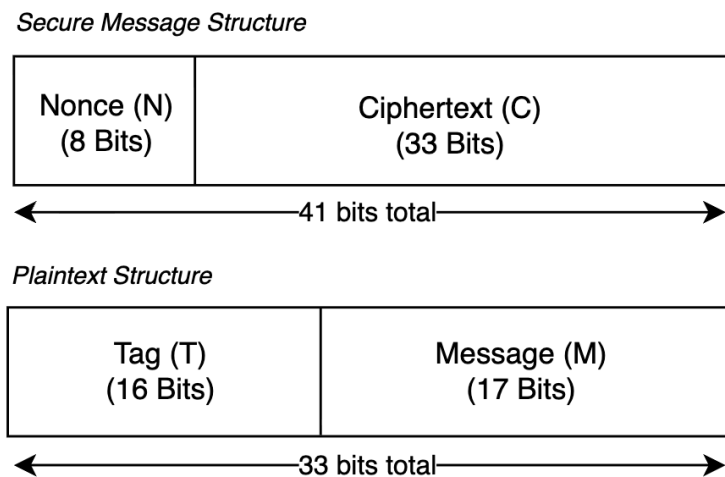
Below, this project proposes a new lightweight cryptographic framework with the protocols reviewed. The notations used to describe the protocol are defined in *Table 4.3* below.

| Notation   | Description                                                    |
|------------|----------------------------------------------------------------|
| Passphrase | An out-of-band, secret shared passphrase between the two users |
| PSS_Key    | Pre-Shared Secret key                                          |
| PBKDF2     | Password-Based Key Derivation Function version 2               |
| M          | Original data frame of the payment message                     |
| P          | Plaintext combines the message and the tag                     |
| C          | Ciphertext that encrypted the plaintext                        |
| Payload    | The final data stream transmitted via vibration                |
| N          | Nonce                                                          |
| T          | The authentication Tag                                         |
| K          | The pseudorandom Keystream generated by AES-CTR                |

|             |                                                      |
|-------------|------------------------------------------------------|
| HMAC        | Hash-based Message Authentication Code               |
| truncate    | A standard function that shortens a binary bitstring |
| AES_CTR     | AES in Counter mode                                  |
| $\oplus$    | The bitwise XOR (Exclusive OR) operation             |
| $\parallel$ | A concatenation operation                            |

*Table 4.3: Notations*

The original 17-bit data will not be transmitted at first. It is encrypted and wrapped into a new secure data. The new total data transmitted over the vibration channel is 41 bits, including the security measures. *Figure 4.16* below shows the new data structure after encryption, which will be sent to the receiver, along with the details of the modified plaintext.



*Figure 4.16: Final Data Stream Structure*

An 8-bit nonce (N) is a random initialization vector included in the message to ensure that each transaction is unique. The system will generate a random 8-bit binary string from 0 to 255 and use it as the nonce, in line with the OOK scheme for the vibration channel. Inside the 33-bit plaintext, there is a 16-bit HMAC tag (T), truncated from HMAC-SHA256, and the original 17-bit vibration message (M).



A general workflow of the proposed schemes is detailed as follows:

**First step:** Both the sender and Receiver verbally agree on a passphrase, and the system will derive an identical 128-bit PSS Key for both users.

$$- PSS\_Key = PBKDF2(passphrase)$$

**Second step:** The sender will first construct a 17-bit message for the payment request and generate a unique 8-bit nonce. Then, the system will generate a 16-bit HMAC integrity tag using the message, nonce, and the PSS key, and also generate a 33-bit AES-CTR keystream using the PSS key and Nonce. Next, the system combines the message and the tag into plaintext and encrypts it using the keystream via the XOR operation to produce the ciphertext. Finally, the system assembles the nonce and ciphertext into a 41-bit payload and passes it to the vibration controller for transmission.

$$- T = truncate(HMAC(PSS\_Key, M || N), 16 \text{ bits})$$

$$- K = AES\_CTR(PSS\_Key, N)$$

$$- P = M || T$$

$$- C = P \oplus K$$

$$- Payload = N || C$$

**Third step:** It is the reverse of the second step, where the Vibration Decoder reconstructs the 41-bit stream from the vibrations and splits the received payload into an 8-bit nonce and a 33-bit ciphertext. Then the system uses the PSS key and the received nonce to generate the same keystream and decrypt the ciphertext back to the plaintext. Next, the system will split the 33-bit plaintext into the received 16-bit tag and the 17-bit messages, and calculate its own expected tag for verification. If the expected tag matches the received tag, it indicates the data's authentication and integrity, and the transaction will be processed. Otherwise, the transaction will be terminated by the protocol.

$$- K = AES\_CTR(PSS\_Key, N')$$

$$- P' = C' \oplus K$$

$$- T' = truncate(HMAC(PSS\_Key, M' || N'), 16 \text{ bits})$$

Figure 4.17 below shows the description of the proposed protocol.

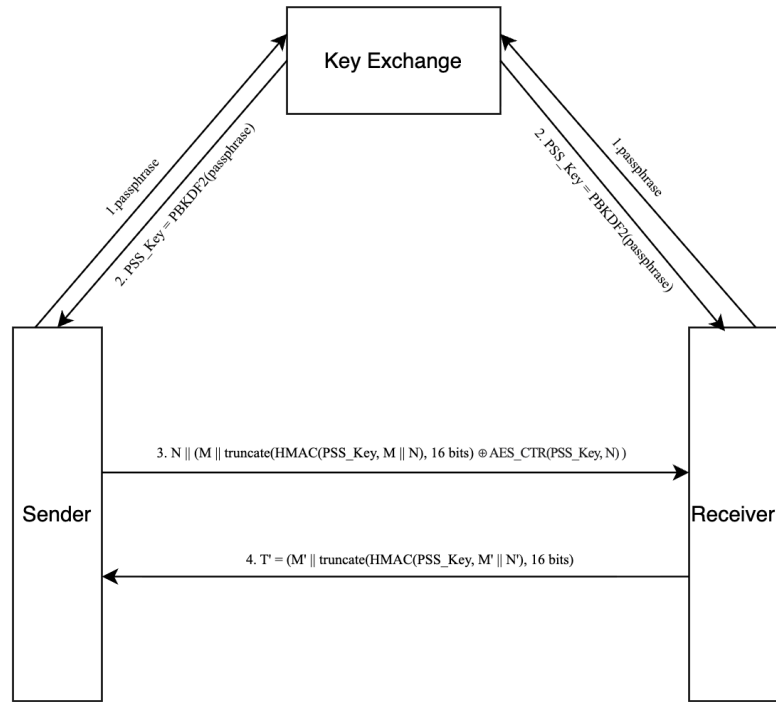


Figure 4.17: Description of the proposed protocol

The figures below show one typical operation performed behind LWC during sending and receiving vibration transactions. Figure 4.18 shows how the PSS key and encrypted message are established from the sender, and Figure 4.19 demonstrates how the receiver decrypts the payload and decodes the vibration patterns into the payment command.

```

=== Secure Sender Build Start ===
M (17-bit)=00010110000100011
Nonce N (0..255)=183
N (8-bit)=10110111
T (16-bit, truncated HMAC-SHA256)=0101110110001111
P = M || T (33-bit)=000101100001000110101110110001111
K (33-bit keystream via AES-CTR)=0111010101101011010001001001001
C = P XOR K (33-bit)=01100011011110101111111111000110
Payload = N || C (41-bit)=101101110110001101111010111111111000110
=== Secure Sender Build End ===
    
```

Figure 4.18: Establishment of the PSS key and secure payload

```

[35725] COMPLETED BIT DETECTION: 41 bits
[35725] RAW BINARY DETECTED: 1011011101100011011110101111111111000110 (41 bits)
rawBits(len=41)=1011011101100011011110101111111111000110
[35725] SECURE RX: payloadBits41=1011011101100011011110101111111111000110
[35725] SECURE RX: N(bits)=10110111 N(int)=183
[35725] SECURE RX: C(33)=01100011011110101111111111000110
secureRx payload41=1011011101100011011110101111111111000110
secureRx nonceBits8=10110111 nonce=183
secureRx ciphertext33=01100011011110101111111111000110
[35726] SECURE RX: K(33)=011101010110101101010001001001001
[35726] SECURE RX: P'(33)=C XOR K =000101100001000110101110110001111
secureRx keystream33=011101010110101101010001001001001
secureRx plaintext33=000101100001000110101110110001111
[35726] SECURE RX: M'(17)=00010110000100011
[35726] SECURE RX: T_recv(16)=0101110110001111
[35726] SECURE RX: T_exp(16)=0101110110001111
secureRx tagRecv=0101110110001111 tagExp=0101110110001111
[35727] SECURE RX: TAG OK → decode legacy 17-bit message
[35727] FINAL DECODE: Searching for pattern in 17 bits
[35727] FOUND COMMAND: 'a' at index=0 (ASCII=97)
[35727] SUCCESS: Single command 'a' decoded

```

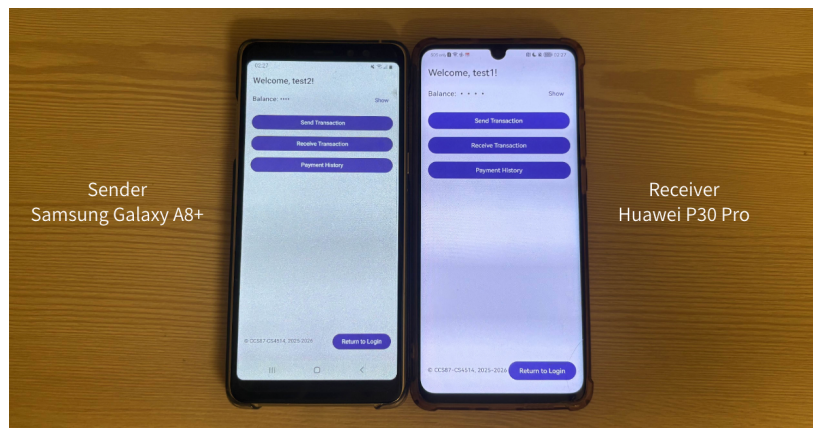
*Figure 4.19: Procedures of the decryption scheme*

## 5. Testing Results and Discussion

As mentioned before, this project addresses the limitations of using the vibration channel in the offline P2P payment system and designs a lightweight cryptographic protocol for this channel to meet security goals. In this section, we will present the findings from the system's functional, performance and penetration testing. Moreover, we will discuss the possible reasons for each outcome and whether each outcome fulfilled the project goals.

### 5.1 System Functional Testing Results

As mentioned in section 3.4, the results will be illustrated by presenting functional test cases for different scenarios that may be encountered when using the application. The primary objective of this phase was to validate the functional logic of the proposed secure P2P vibration payment system. *Figure 5.1* shows the controlled test environment setup, where the sender and receiver are closely attached to each other and are placed on a stable wooden table. Appendix C presents the use cases performed during system functional testing and the final results of each test.



*Figure 5.1: Users' test environment setup*

Overall, the successful results of all use cases demonstrated that the P2P payment system of the software-to-hardware pipeline is correctly implemented. From the usability perspective, it shows that a user can perform the application as a financial application. The sender can encode and encrypt the transaction request into a vibration pattern, and the receiver recognizes and decrypts it to the payment command. This proves that the proposed vibration communication channel and lightweight cryptography protocol are logically functional and usable in a secure P2P payment system background.

However, we should note that these tests were conducted at 0 cm, which represents the "ideal environment" where vibration energy loss is minimal. They have not yet proven the system's

resilience to external variables such as communication distance or surface materials. Moreover, we did not test the performance of each magnitude threshold of our decode algorithm. It indicated that there are areas for further investigation to prove that it is a comprehensive secure P2P payment system.

## 5.2 Performance Testing Results

To address the issues stated in the previous subsection, we also test the accuracy of the designed vibration-encode and decode algorithm at different adaptive magnitude thresholds, communication distances & direction, and surface materials, both with and without the cryptographic framework. The sender and receiver are still the same mobile phones presented in section 5.1.

### 5.2.1 Performance based on Magnitude-Based Threshold

Vibration detection is tested with different thresholds by varying the magnitude required to be recognised as vibration, since the idle-tested phone magnitude ranges from 9.59 to 9.61. Therefore, the tested magnitude thresholds should be higher than the range value, such as 9.65, 9.75, 9.8, 9.9, and 10. The accuracy of the received data reflects its correctness.

| Threshold | Without LWC Framework | With LWC Framework |
|-----------|-----------------------|--------------------|
| 9.65      | >95%                  | >95%               |
| 9.75      | >95%                  | >95%               |
| 9.8       | ~77%                  | ~54%               |
| 9.9       | ~65%                  | ~49%               |
| 10        | ~65%                  | ~36%               |

*Table 5.1: Performance based on magnitude threshold*

Table 5.1 shows the results. It can be observed that accuracy decreases significantly after the threshold exceeds 9.75; it continues to drop thereafter, and the transaction will fail. Furthermore, we spotted that the accuracy without the LWC framework was fixed around 65%, whereas with the LWC framework it performed poorly and unstably. The possible reason for these interesting results is that the transaction without the LWC framework generated the same vibration pattern according to the defined algorithm; the results are always the same, and the receiver may not be able to map the same bit in every attempt. But if the algorithm uses the LWC framework, since the nonce introduces randomness and generates different vibration patterns in every transaction after encryption, the structure is more complex, leading to vibration data that may not map to the

receiver's arbitrary state. However, if the threshold is within 9.65-9.75, the accuracy remains high and stable, and the transaction will succeed regardless of whether the LWC framework is used. It shows that the ideal threshold for the transaction lies between 9.65 and 9.75.

### 5.2.2 Performance based on Medium Objects

Different types of media used for transactions may also affect accuracy. We tested our vibration channel on four common types of medium objects. The experimental setup is also similar to section 5.1, but the placement locations are different, and the same ideal threshold is used.

| Materials       | Without LWC Framework | With LWC Framework |
|-----------------|-----------------------|--------------------|
| Wood Table      | >95%                  | >95%               |
| Metal Shelf     | >95%                  | >95%               |
| Cushioned chair | ~85%                  | ~68%               |
| Cotton bed      | ~65%                  | ~51%               |

*Table 5.2: Performance based on surface materials*

From Table 5.2, we see that transactions on the wood table and the metal shelf achieve higher accuracy. In comparison, the cushioned chair and cotton bed have lower accuracy, leading to failed transactions. The possible reason for this result is the material density. Wood and table are more solid and high-density, with low acoustic damping, resulting in better performance. However, the cushioned chair and cotton are more porous; the soft materials may absorb and weaken high-frequency vibration, making it difficult for the receiver to decode the data.

### 5.2.3 Performance based on Communication Distance

The communication distance may also affect the accuracy, as it is a close-proximity channel; the vibration strength may get weaker for the receiver to detect. We should also test the data reception at distances of 10cm, 20cm, 30cm, and 40cm.

| Distance | Without LWC Framework | With LWC Framework |
|----------|-----------------------|--------------------|
| 10cm     | >95%                  | >95%               |
| 20cm     | >95%                  | >95%               |
| 30cm     | ~65%                  | ~41%               |
| 40cm     | ~65%                  | ~32%               |

*Table 5.3: Performance based on transaction distance*

Table 5.3 presents the testing results. We saw that accuracy becomes low and fixed after exceeding 30cm, indicating that the receiver could not receive all the messages correctly beyond that distance, and all transactions will fail thereafter. It has also been shown that the effective range of the transaction may be within 20cm, and our proposed secure vibration channel is robust for transferring vibration patterns within this range.

#### 5.2.4 Performance based on Different Communication Directions

We have also tested our transactions in different directions, as different angles may affect accuracy. Figure 5.2 below shows the overall settings (a) and the final performance results (b).

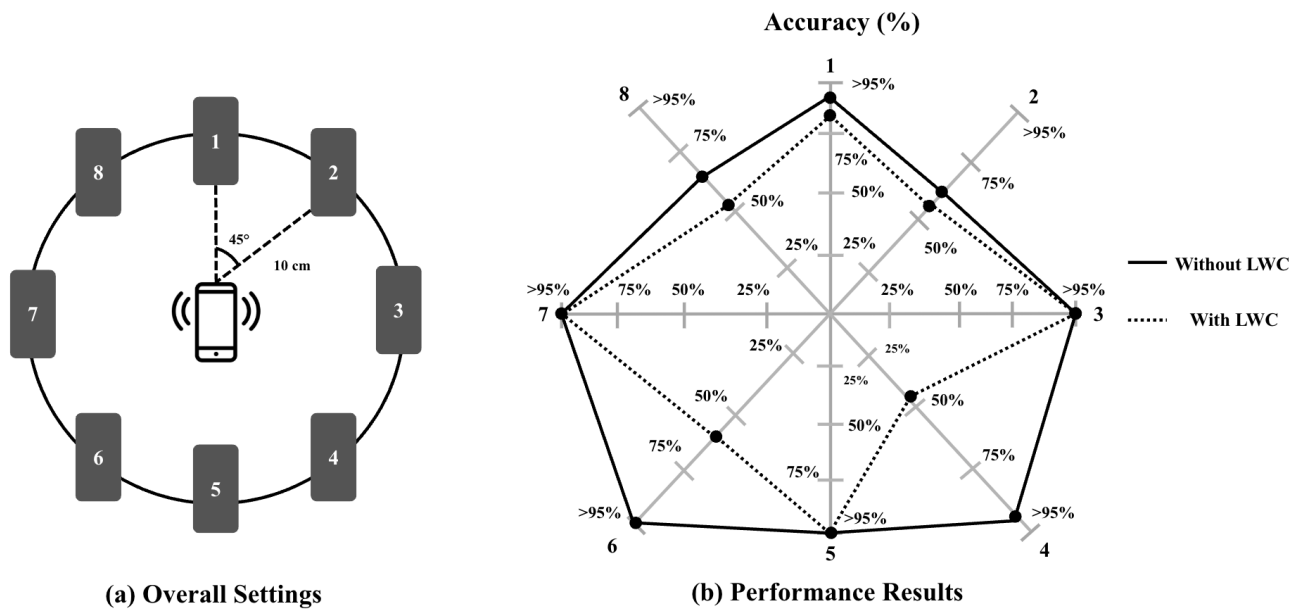


Figure 5.2: Test settings & Performance based on different communication directions

The results show that the algorithm performed best when it was in a horizontal position (Positions 3 & 7) and when the sender was at the top of the receiver (Position 5). This suggests that the smartphone's built-in vibrator and accelerometer do not emit energy uniformly in all directions, and that internal hardware placement and the device casing likely create attenuated paths. In our experimental setup, the sender vibration motor is at the bottom of the device, and the receiver accelerometer is at the top of the phone. This may explain why it performs better in the suggested result position. Therefore, to ensure the transaction's success, the device must be aligned with the target position with greater accuracy, and results may differ if another mobile device is used.

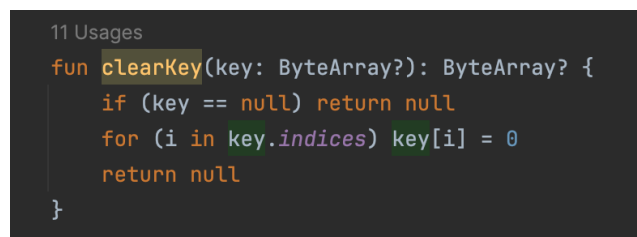
### 5.2.5 Discussion Based on Performance Results

To conclude, results from section 5.2 have demonstrated that our proposed secure P2P payment system using the vibration channel may be viable and reliable under certain circumstances, such as when the vibration algorithm is under an ideal attitude threshold, it is placed on a stable medium object for transaction, in a short communication distance, and in a horizontal direction setting.

### 5.3 Penetration Testing Results

The pen test will focus on the security of the vibration communication channel and the key establishment. Since the key is used to encrypt the vibration data in each transaction, it is expected that the key will not be reused. Hence, the ethical hacking attack is expected to mainly target the proposed security design to test whether it is vulnerable to attacks such as replay attacks and eavesdropping.

First, to address key reuse, our implementation of the LWC protocol uses the clearkey function after the sender and receiver have completed the transaction, the transaction timed out, or the transaction was cancelled (see *Figure 5.3* below). It ensures the key won't be reused and remains unique for encryption.

A screenshot of a code editor showing the implementation of the `clearKey` function in Kotlin. The code is as follows:

```
11 Usages
fun clearKey(key: ByteArray?): ByteArray? {
    if (key == null) return null
    for (i in key.indices) key[i] = 0
    return null
}
```

*Figure 5.3: ClearKey function*

Next, it is essential to examine the security challenges and highlight the necessity of a cryptographic design. First, it may be vulnerable to eavesdropping if the attacker is also in physical proximity and could intercept the vibrations, decode the messages, and discover the payment request from the users. Next, it is possible to replay attacks, as an attacker can record a valid vibration pattern and replay it later to authorize a duplicate payment.

In our design, the sender and receiver share a passphrase to derive a unique 16-byte PSS key; the system then generates a 33-bit keystream using AES in Counter (CTR) mode with the PSS key and an 8-bit unique nonce. Lastly, the 17-bit payment message is combined with a 16-bit HMAC tag and then XORed with the keystream; a 44-bit payload is sent and received. Consequently, the security results show that even if the attacker is in physical proximity and intercepts the vibration



pattern, they cannot decode our payload without the secret PSS key, which only the original sender and receiver know, keeping the payment details confidential. Moreover, if the attacker records and replays the old vibration message, the receiver will detect that the nonce or the resulting HMAC tag is no longer valid and terminate the transaction. The double protocol protects the transaction integrity.

Moreover, the results of the performance-based test on communication distance indicate that the scammer needs to be at least 20cm away to eavesdrop. The proposed secure transaction channel ensures that two users are in proximity and that attacks are improbable, as users can easily detect third parties and identify the sender and receiver in a real-life setting, thereby guaranteeing the security of the payment channel.

Furthermore, we analyzed our designed system from the perspective of a secure mobile payment system; we have proof that our system is safe in terms of confidentiality and integrity, based on the prevention of the attack described above. Our system also provides authentication using a secret (passphrase & PSS key) known only to authorized parties, and the received HMAC tag confirms the origin of the vibration data. Then, the unique nonce and the PSS Key create a verifiable link to a specific transaction that the user cannot deny. Access control is also satisfied, as only users who know their account details can log in, and users without an account cannot access the payment system.

## **6. Conclusion**

### **6.1 Summary of Achievement**

This project introduced a secure alternative to traditional P2P mobile payments by replacing vulnerable Radio-frequency (RF) signals with a radio-free vibration communication channel.

To start with, this project has developed a simple interface and database for the mobile payment system. Initially, users could register for accounts, log in, and perform simple functions such as transactions and checking payment history, like a financial application. We have further developed a P2P payment system based on this prototype by introducing a novel physical medium that grips the smartphone's internal hardware. We used the vibrator and accelerometer as transmitter and receiver, respectively, to facilitate data transfer via physical contact via vibration patterns and facilitate the proposed P2P transaction medium, for which previous research has been less in-depth.

Moreover, we included a custom, lightweight cryptography (LWC) framework that optimizes security for a low-speed data-stream communication channel. This also lacks prior research, and few researchers have attempted to implement security measures directly in vibration communication channels, especially using a lightweight approach. The proposed LWC framework is a creative hybrid security design that combines out-of-band key establishment via a shared passphrase with AES-CTR encryption and HMAC-SHA256 integrity tags to meet the security requirements of a mobile payment system.

Lastly, we have performed functional, performance, and penetration testing to ensure the system's usability and reliability, and we also hope to encourage further research in related fields to develop more creative designs for this low-cost, low-energy-use approach.

### **6.2 Critical Review**

Review the whole project: We figured out that the simplest task is to build the Android mobile payment system's interface and incorporate the LWC framework, and the most complicated is to implement the vibration communication channel.

The UI design is simple to design as the writer has taken related design courses during undergraduate studies, and modern development tools and Android's well-documented SDK provide out-of-the-box automation components. Moreover, incorporating the LWC framework

was straightforward because the underlying physical vibration channel was already fully functional by that stage, and many well-established cryptographic libraries are available for Android, allowing us to directly integrate standard protocols into the existing transaction flow without building the encryption scheme from scratch.

For the vibration communication channel, we found implementing it challenging, as many situations were not encountered during the design phase. For example, due to real time delays of starting the transaction, the decoding algorithm may sometimes incorrectly decode the command, for instance, Sometimes when the decoder tries to decode the character "b," it may confuse it with the character "c," since the 7-bits binary representations of both characters are similar: 1100010 ("b") and 1100011 ("c"). The algorithm may incorrectly recognize the last bit as "1" but not "0" and make a wrong decision. The reason for this fault is probably that the algorithm recognizes the high vibration in the previous window in the current window, which misleads the last bit. To solve the problem, we have to repeatedly test the algorithm and carefully adjust the threshold and sample count, so that even when some samples are included in the wrong windows, they won't meet the algorithm's condition and won't be recognized as "1".

### **6.3 Further Work**

The proposed system could be further extended to improve the performance of the vibration channel for transactions and fulfill the security goals of the mobile payment system. First, consider that, in the current hardware approach, a more reliable accelerometer could be used to prevent noise when collecting data, such as a microphone and gyroscope, which offer better high-frequency detection and capture of micro-directions through sensor fusion, helping to reduce noise.

Nevertheless, the LWC framework performs worse and is less stable than the one without it across all performance results. This may be because the payload to be transferred has increased; we suggest improving the current decoding algorithm by introducing additional techniques, such as adaptive amplification and frequency adjustment, to enhance the accuracy of the proposed LWC framework.

Lastly, we identify a security area that may require improvement: the current protocol may not be secure from an availability perspective, as an attacker could inject a random vibration pattern to disrupt the transaction process. Further work may implement a collision-detection mechanism in the decode algorithm to mitigate interference from unknown sources.

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## Appendices

### Appendix A - Monthly Log

#### October 2025

I will provide my progress report in three sections each month.

It includes: 1) Documentation 2) Implementation 3) Expectations/ goals for the next month.

For Documentation, I have reviewed the comments on the project plan from my supervisor and have completed the draft of the introduction and literature review part for the interim report I this month. Moreover, I have brainstormed the design and testing methods for the overall system.

For implementation, I have completed the UI coding part of the P2P payment system. However, I'm kind of behind schedule due to the busy midterm weeks and assignments. For the vibration channel part, I have completed the encoding scheme to generate vibration. However, I am still stuck on a minor issue related to detecting vibration by calculating magnitude data for the decoding scheme.

Expectations to be complete next month:

- Complete the draft of the proposed design and implementation/testing schedule of the interim report 1 and submit it to my supervisor for suggestions and reviews
- Complete basic vibration communication channel code development (Without encryption)
- Study on proposed lightweight cryptographic protocols and consider how to integrate them into the communication channel

#### November 2025

For Documentation, I have completed the remain part of the interim report I and submitted to my supervisor. We have also scheduled a meeting to talk about the project. He have provided some suggestion on the implementation method of the vibration design and solved my questions.

For implementation, I have completed to remain decoding part and start considering how to integrate the cryptographic protocols into the communication channel.

Expectations to be complete next month:

- Study on the proposed lightweight cryptographic protocols and start implementing it into the vibration communication channel
- Check the comment from my supervisor on interim report I and start consider how to write interm report II

#### December 2025

For Documentation, I reviewed the comments from my supervisor and modified the requirements that he required me to change and started writing my implementation details on the interim report II

For implementation, I have completed the decoding part; however, some issues have been identified. I have found some time to address the issue, and now consider the design of incorporating security protocols into it.

Expectations to be complete next month:

- Finish the proposed lightweight cryptographic protocols in the vibration communication channel
- Extract the initial testing results on the design and complete the interim report II

## **January 2026**

For implementation, after some minor fix of UI and vibration communication channel, I have successfully integrated the purposed cryptography inside the P2P payment system, I'm currently evaluating the developed system from test cases and performances and extract the initial testing results.

For Documentation, I have completed the interim report 2, and send to my supervisor for reviewing. I'm waiting for his comments and prepare for the final submissions.

Expectations to be complete next month:

- Improve the quality of the sources code and identify any hidden bugs
- Continue testing the developed system
- Preparation for the final report and submission required

## **February 2026**

For implementation, I have completed testing the entire system and extracted the final results for the usability, performance, and security tests. Next, I improved the source code quality by restructuring and adopting design patterns.

For Documentation, I have reviewed the comments on the interim report II from my supervisor and preparing (Some writing) the final report for the last submission. I have also checked the guidelines for other required deliverables, such as presentations and demonstrations.

Expectations to be complete next month:

- Complete the final report of the project
- Prepare for the presentation and demonstration of the project

## **March 2026**

For implementation, Since this is the last month of the project, all the focus is shifted to documentation, there is no more implementation progress as the coding part is completed.

For Documentation, I have completed:

- The final report
- Poster for CS Final Year Project Exhibition 2026
- Presentation slides

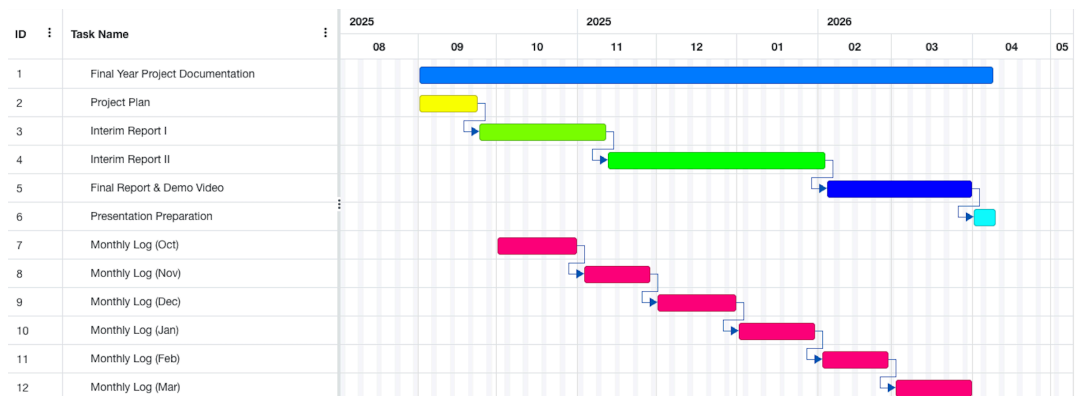
Expectations to be complete next month (First week):

- Screenshoot & Demonstration Video

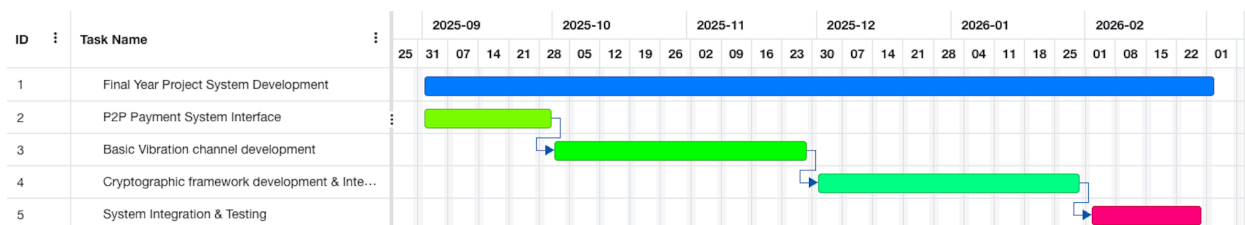
## Appendix B - High-level project schedule plan & Gantt Chart

| Goals                                                                                                         | Key Deliverables                                                                                                       | Improvement on the previous phase                                                                    | Testing                                                                                                               | Documentation                                |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Phase 1: P2P Payment System Interface (September 2025)</b>                                                 |                                                                                                                        |                                                                                                      |                                                                                                                       |                                              |
| Develop a P2P payment system interface for two Android phones                                                 | A functional payment interface with basic transaction operations                                                       | N/A                                                                                                  | Unit testing on UI interfaces                                                                                         | Project Plan                                 |
| <b>Phase 2: Basic Vibration channel development (October – November 2025)</b>                                 |                                                                                                                        |                                                                                                      |                                                                                                                       |                                              |
| Set up the development environment and implement core vibration transmission and reception between two phones | A basic Android app that can encode a payment command into a vibration frame and decode it back on the receiver device | Implementing vibration and payment request into the UI interface                                     | Unit and integration testing on the UI interfaces and the vibration communication channel                             | Interim Report I                             |
| <b>Phase 3: Cryptographic framework development &amp; Integration (December 2025 – January 2026)</b>          |                                                                                                                        |                                                                                                      |                                                                                                                       |                                              |
| Design and develop the passphrase-based key exchange and AES encryption with the HMAC scheme                  | An Android app prototype that can perform a secure key exchange and send an encrypted message via vibration            | Integrating the proposed secure protocols into the vibration communication channel and UI interfaces | Unit and integration testing on the UI interfaces and the vibration communication with secure cryptographic protocols | Interim Report II preparation                |
| <b>Phase 4: System Integration &amp; Testing (February 2026)</b>                                              |                                                                                                                        |                                                                                                      |                                                                                                                       |                                              |
| Integrate all components into a single, comprehensive system and conduct testing and evaluation of the system | A fully functional end-to-end secure P2P vibration mobile payment application                                          | Improve the readability and quality of the source code                                               | System testing on the whole system, and perform preliminary performance and Penetration testing                       | Interim Report II & Final Report preparation |
| <b>Phase 5: Final Preparation &amp; Documentation (March 2026)</b>                                            |                                                                                                                        |                                                                                                      |                                                                                                                       |                                              |
| Prepare the final report, create a demo video, and fine-tune the source code for                              | Final project report, complete source code, demonstration                                                              | Improve the readability and quality of the source code                                               | Final performance and penetration testing                                                                             | Same as Key Deliverables                     |

|                                                                                |                                    |  |  |  |
|--------------------------------------------------------------------------------|------------------------------------|--|--|--|
| final submission.<br><br>Moreover,<br>preparation for the<br>FYP presentation. | video, and a<br>presentation slide |  |  |  |
|--------------------------------------------------------------------------------|------------------------------------|--|--|--|



*A Gantt Chart for the Documentation Part of the Project*



*A Gantt Chart for the System Development part of the Project*

## Appendix C - Use cases and the final functional testing results

| <u><b>User Create Account</b></u>                                                                                 |                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Requirement: User type the username and a password longer than 6 character for register account                   |                                                                                                                            |
| Test Case:                                                                                                        | Expected Result:                                                                                                           |
| User input a valid username and valid password<br>Username is not being registered                                | User created an account in local database and forward back to the login page for login                                     |
| User input a valid username and valid password<br>But the username is being registered                            | Alert ("Username is being registered, please use another name!")<br>Stay in register page                                  |
| User input a invalid username (Exceed String Limit or too short) or invalid password that break the security rule | Alert ("Invalid username, Please try again!") or<br>Alert ("Invalid password, Please try again!")<br>Stay in register page |
| <u><b>Final Results: Success</b></u>                                                                              |                                                                                                                            |

| <u><b>User login in account</b></u>                                |
|--------------------------------------------------------------------|
| Requirement: User type the valid username and password to login in |

| Test Case:                                                             | Expected Result:                                                                                                |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| User input a valid username and valid password<br>(Account is existed) | User Logged in, forward user to Home menu display<br>welcome message<br>User data retrieved from local database |
| User input a invalid username and valid password                       | Alert ("Please input an valid username!")<br>Stay in login page                                                 |
| User input a valid username and invalid password                       | Alert ("Invalid password, Please try again!")<br>Stay in login page                                             |
| User input invalid username or password multiple time                  | Alert ("Login in function locked, please try again<br>later!")<br>Back to mobile home page                      |
| <b><u>Final Results: Success</u></b>                                   |                                                                                                                 |

| <b><u>Checking payment history</u></b>                                            |                                                          |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|
| Requirement: User want to check transaction record and enter payment history page |                                                          |
| Test Case:                                                                        | Expected Result:                                         |
| User has payment record                                                           | Automatically display the transaction record in the page |
| User has no payment record                                                        | No payment record in the page                            |
| <b><u>Final Results: Success</u></b>                                              |                                                          |

| <b><u>Admin Management page</u></b>                                                       |                                                    |
|-------------------------------------------------------------------------------------------|----------------------------------------------------|
| Requirement: Administrator want to check the transaction records and users account status |                                                    |
| Test Case:                                                                                | Expected Result:                                   |
| Admin click "User Account Status"                                                         | Display all the user (Including admin) in the list |
| Admin click "Transaction Log"                                                             | Display all user transaction record                |
| <b><u>Final Results: Success</u></b>                                                      |                                                    |

| <b><u>PSS Key generation</u></b>                                                                                 |                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Requirement: User's establish the share key using passphrase before performing transaction (Sender and Receiver) |                                                                                                                                               |
| Test Case:                                                                                                       | Expected Result:                                                                                                                              |
| Both user's enter the exact same secret passphrase                                                               | Both devices process the passphrase and generate the identical PSS key for processing transaction later                                       |
| One of the user enter the different secret passphrase                                                            | The generated PSS key will be different, transaction will fail alert ("Authentication Failed!" or "Corrupt Data!") during transaction process |
| <b><u>Final Results: Success</u></b>                                                                             |                                                                                                                                               |

|                                        |
|----------------------------------------|
| <b><u>Send Payment Transaction</u></b> |
|----------------------------------------|

|                                                                                                                                                         |                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirement: Sender send payment transaction to receiver (Assume PSS key is successfully establish for both users and place phones on the same surface) |                                                                                                                                                                          |
| Test Case:                                                                                                                                              | Expected Result:                                                                                                                                                         |
| Tap “Send Payment”                                                                                                                                      | The payment request will be encrypted and the vibration encoder will encode the data based on the OOK algorithm to vibration pattern. The phone will physically vibrates |
| User tap “Cancel payment request” after requesting payment                                                                                              | The transaction request will be terminated, no money will be paid                                                                                                        |
| <b><u>Final Results: Success</u></b>                                                                                                                    |                                                                                                                                                                          |

|                                                                                                                                                            |                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Receive Payment Transaction</u></b>                                                                                                                  |                                                                                                                                                                                                      |
| Requirement: Receiver receive payment transaction to sender (Assume PSS key is successfully establish for both users and place phones on the same surface) |                                                                                                                                                                                                      |
| Test Case:                                                                                                                                                 | Expected Result:                                                                                                                                                                                     |
| Tap “Receive Payment” (HMAC Tag same)                                                                                                                      | The vibration decoder will listen and capture the vibration pattern and and decodes it using the adaptive threshold algorithm, decrypt the data, calculate the expected tag and confirms the payment |
| Tap “Receive Payment” (HMAC Tag not the same)                                                                                                              | After calculatd the expected tag but it is not same with the received tag, transaction will fail and terminated, alert ("Authentication Failed!")                                                    |
| User tap “Cancel payment receive” after receive payment                                                                                                    | The listen of vibration will be terminated, no money will be receive                                                                                                                                 |
| <b><u>Final Results: Success</u></b>                                                                                                                       |                                                                                                                                                                                                      |